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Retrieval of orbital maneuver information of Starlink satellites from SpaceX-released ephemerides by analytical and numerical methods

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Abstract

Starlink satellites have caused widespread concern in the fields of astronomical observation and space situational awareness, as they are too numerous and their orbital states are relatively complex. It requires in-depth understanding and knowledge of fundamental information of orbital maneuvers of the Starlink satellites during their in-orbit operations to help improving the accuracy of orbit determination and prediction. SpaceX-released ephemerides contain predicted maneuvers of the Starlink satellites. Taking the orbital semi-major axis as a characteristic parameter for detection of orbital maneuvers, analytical and numerical methods are proposed to retrieve the orbital maneuvers of the Starlink satellites from the SpaceX-released ephemerides, and the fundamental maneuver information is investigated with examples in this paper. Using the ephemerides released by the SpaceX from September 1st to 5th, 2024 (UTC) for the selected Starlink satellites in the operational, parking, orbit raising and deorbit phases respectively, the time series of the orbital semi-major axis variations per unit time (1 min) caused by both the maneuver force and the atmosphere drag (which can be approximated as the maneuver force) are retrieved by the analytical method; and the corresponding time series of the orbital semi-major axis variations per unit time (1 min) caused solely by the maneuver force are also retrieved by the numerical method. Then, the fundamental information of orbital maneuvers of the selected Starlink satellites is revealed intuitively.
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Keywords: Starlink satellite; Orbital maneuver; SpaceX-released ephemeris; Analytical retrieval; Numerical retrieval

1. Introduction

With increasingly human space activities, the numbers of in-orbit payloads, rocket bodies and space debris are rapidly growing. As of May 8th, 2025, the total number of historical cataloging payloads by Space-Track is 20,540, of which there are as many as 8,379 Starlink satellites[2], and these numbers will continue to grow. The Starlink satellite missions involve the parking, orbit raising, operational and deorbit phases, coupled with frequent maneuvers, resulting in relatively complex orbital states. Given the massive scale and relatively complex orbital states, Starlink satellite constellation poses potential threats to in-orbit spacecrafts (Li et al., 2022). It is essential to acquire timely and accurate orbit information of the Starlink satellites to reduce or avoid their potential threats

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[2] https://www.space-track.org/#catalog (accessed on May 8th, 2025)

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effectively. SpaceX is providing ephemeris[3] (orbit information) that can be used for conjunction assessment purposes. However, the Starlink satellites frequently perform orbital maneuvers during their in-orbit operations for various needs (Liu et al., 2024b); the SpaceX-released ephemerides do not contain maneuver information directly. In the case of unknown satellite orbital maneuver information, it is difficult to accurately model the maneuver forces, which restricts the accuracy of orbit determination and prediction significantly. There is no doubt about the in-depth understanding and knowledge of the fundamental information of orbital maneuvers of the Starlink satellites during their in-orbit operations to bring positive effects on the orbit determination and prediction, etc.

Starting from May 2021, the SpaceX releases and updates predicted ephemerides of the Starlink satellites at a frequency of 3 times a day, approximately every 8 h[4]. They contain a time series of state vectors obtained through a propagation with maneuvers. In other words, these ephemerides contain predicted maneuvers. Thus, the orbital maneuvers can be inferred from them. The SpaceX-released ephemerides provide reliable data resources to explore the fundamental maneuver information of the Starlink satellites during the in-orbit operations (Zhang and Shen, 2022; Liu et al., 2024a).

Since the gradual implementation of the SpaceX Starlink mega constellation, the Starlink satellites have caused widespread concern in the fields of astronomical observation (McDowell, 2020; Tyson et al., 2020; Walker et al., 2020; Cui and Xu, 2022; Halferty et al., 2022; Mróz et al., 2022; Shustov, 2022; Di Vruno et al., 2023) and space situational awareness (Alfano et al., 2021; Petit et al., 2021; Reiland et al., 2021; Ashurov, 2022; Li et al., 2022; Zhang and Shen, 2022; Zhang et al., 2022; Burgis et al., 2023; Dutt et al., 2023; Hossein et al., 2023; Tang et al., 2023; Wang and Li, 2023; Baruah et al., 2024; Liu et al., 2024a, 2024b; Ahmad et al., 2025; Aranda et al., 2025; Lin et al., 2025; Liu et al., 2025; Oliveira et al., 2025), etc.

McDowell (2020) assessed the astronomical impact of the Starlink satellite constellation quantitatively by modeling satellite visibility across various latitudes. Walker et al. (2020) conducted comprehensive simulations of the Starlink satellite constellation's impacts on multiple observational modes, and proposed actionable mitigation strategies. Tyson et al. (2020) investigated techniques to mitigate the brightness and trail effects caused by low Earth orbit (LEO) satellites on the Rubin Observatory's Legacy Survey of Space and Time, offering practical strategies to preserve survey quality in wide-field astronomical observations. Mróz et al. (2022) quantified the impact of the Starlink satellites on the Zwicky Transient Facility survey observations, and presented a detailed assessment of how large LEO constellations influence time-domain astronomical surveys. Shustov (2022) discussed the "dark and quiet sky" problem resulting from satellite mega-constellations, and emphasized the broader challenges for astronomical observations associated with large LEO constellation deployments. Cui and Xu (2022) developed two dedicated plugin modules for the WorldWide Telescope platform to simulate and assess the influence of the Starlink satellites on astronomical observations at specific observatories over a given time period. Halferty et al. (2022) presented the photometric characterization and trajectory accuracy of the Starlink satellites, and provided useful insights into their impact on ground-based astronomical surveys. Di Vruno et al. (2023) detected unintended electromagnetic radiation from the Starlink satellites using the LOFAR radio telescope, thereby revealing the potential radio-frequency interference caused by large LEO constellations.

Reiland et al. (2021), Petit et al. (2021), and Alfano et al. (2021) focused on collision risk assessments of the Starlink satellites. Taking the orbital semi-major axis and inclination as characteristic parameters for detection of orbital maneuvers, Ashurov (2022) proposed a maneuver detection method based on the historical Two-Line Element (TLE) data and applied to specific examples for two Starlink satellites. Zhang et al. (2022) investigated the deployment of the Starlink constellation and the collision threats posed by the Starlink satellites in different phases, and pointed out that the accuracy of orbit prediction for the Starlink satellites in the continuous orbit raising and deorbit phases is lower than that in the parking and operational phases; accordingly, the confidence of collision probability for the Starlink satellites in the continuous orbit raising and deorbit phases is also lower than that in the parking and operational phases. Li et al. (2022) achieved accurate tracklet-to-object matching for the Starlink satellites in the orbit raising phase through the recursive orbit determination and prediction, in which an orbit dynamical model incorporating the maneuver force is constructed to improve the accuracy of orbit determination and prediction, the unknown maneuver force is treated as an estimated constant in the process of orbit determination regardless of the essential laws of orbital maneuvers, and the corresponding error of orbit prediction reaches a few kilometers. Zhang and Shen (2022) utilized the SpaceX-released ephemerides of the Starlink satellites to study their orbital characteristics, and inferred that the Starlink satellites possess a fundamental law of periodic "propulsion-pause" in the orbit raising phase. Tang et al. (2023) analyzed the distinctive characteristics of the Starlink satellite constellation, i.e., the motion synchronization of multi-shells and the ground track overlap of satellites in the same shell, and presented concrete steps of orbital design for Starlink-like satellite constellations. Based on the historical TLE data, Wang and Li (2023) applied a machine learning method to the rapid classification of orbital states for the Starlink satellites. Burgis et al. (2023) designed a constellation deployment and orbit raising model, and verified by using the actual TLE data of the Starlink V1.0-L1. Dutt

[3] https://starlink.com/public-files/ephemerides/README.md
[4] https://www.space-track.org/#publicFiles

et al. (2023) examined the effectiveness and applicability of four algorithms on re-entry time prediction of uncontrolled space objects, and applied these algorithms to the re-entry prediction of the Starlink-26 satellite, which is an object selected in the Inter-Agency Space Debris Coordination Committee (IADC) re-entry test campaign. Hossein et al. (2023) characterized the photometric properties of some Starlink satellites through red, green and blue filters. Baruah et al. (2024), Ahmad et al. (2025), and Oliveira et al. (2025) investigated the impact of geomagnetic storms on launch into orbit and re-entry of the Starlink satellites, respectively. Lin et al. (2025) proposed an effective method to compensate orbit prediction errors of the Starlink satellites, and improved the accuracy of orbit prediction significantly. Liu et al. (2025) developed a deep learning-based approach for maneuver forecast of the Starlink satellites. Aranda et al. (2025) explored the collision avoidance through drag augmentation maneuvers, and validated using the Starlink-1043 satellite as a case study.

Recently, Liu et al. (2024a) clarified the basic situation of orbit dynamical model adopted in the SpaceX-released ephemerides through the analysis of mean orbital elements, and applied the SpaceX-released ephemerides to investigate the configuration of the Starlink satellite constellation. Based on the SpaceX-released ephemerides, Liu et al. (2024b) also performed piecewise linear fitting on the time series of the mean orbital semi-major axis, with only the short-period terms of the Earth's gravitational field removed, and combined the corresponding linear fit slopes to detect and extract fundamental strategies of orbital maneuvers of the Starlink satellites.

Based on the research train of thought in Zhang et al. (2018) and Yuan et al. (2024), i.e., the retrieval of the orbital semi-major axis variations per unit time of re-entry objects caused by the atmospheric drag from precise orbit data, the appropriate amendments to the orbital maneuver extraction method proposed by Liu et al. (2024b) are made, and analytical and numerical methods are presented to retrieve orbital maneuvers of the Starlink satellites from the SpaceX-released ephemerides in this paper, by taking the orbital semi-major axis as a characteristic parameter for detection of orbital maneuvers. Moreover, using the ephemerides released by the SpaceX from September 1st to 5th, 2024 (UTC), the fundamental information of orbital maneuvers of the selected Starlink satellites in the operational, parking, orbit raising and deorbit phases are retrieved successfully.

2. Retrieval methods

The orbital motion of satellites around the Earth is subjected to various dynamic factors, including natural forces and maneuver forces. The satellite orbit is determined by the combined effects of these forces. Evidently, the SpaceX-released ephemerides contain the effects of main natural forces on the orbital motion of the Starlink satellites, especially the Earth's nonspherical gravitational force and the atmospheric drag, as well as the effect of the maneuver forces. In orbital maneuver discriminations, the orbital semi-major axis can be served as an important characteristic parameter (Liu et al., 2024b; Ashurov, 2022). Therefore, based on the SpaceX-released ephemerides, if the effects of the main natural forces are subtracted from the original semi-major axis of the Starlink satellites, the effect of the maneuver forces can be obtained more clearly and accurately. Consequently, orbital maneuvers can be effectively retrieved from the orbital semi-major axis variations, and fundamental information of orbital maneuvers are ultimately acquired by statistical analysis.

Similar to Zhang et al. (2018), the effects of the main natural forces can be subtracted from the original semi-major axis with two different approaches, i.e., analytical method and numerical method. Accordingly, basic principles and computational procedure of the analytical and numerical methods to retrieve the orbital maneuver information of the Starlink satellites from the SpaceX-released ephemerides are elaborated in the following.

It is worth noting that the SpaceX releases and updates predicted ephemerides of the Starlink satellites at a frequency of 3 times a day, approximately every 8 h, and the ephemerides in the files released at adjacent times have overlapping time coverage (overlapping segments). The more recently released segment (newer data) is expected to be more accurate. Therefore, for the overlapping segments, only the latest released data (the data from the later or newer file) for the Starlink satellites are retained and spliced, by using the start time of the latest segment as the endpoint of the previous one, to form the final ephemeris data used for analysis. For the stitching of ephemeris data of the Starlink satellites released by the SpaceX from September 1st to 5th, 2024 (UTC), which will be utilized for subsequent analysis, the ephemeris files released for the first time on September 6th, 2024 (UTC) are also involved to ensure the full dataset is accurate. The start times of the ephemerides extracted from these files are then used as the endpoints for the ephemeris data released for the third time on September 5th, 2024 (UTC). Note that the final ephemeris data formed from these ephemerides released by the SpaceX from September 1st to 5th, 2024 (UTC) incorporate the partial ephemerides of the Starlink satellites on September 6th, 2024 (UTC).

2.1. Basic principles

2.1.1. Analytical method

According to the analytical theories for satellite orbital perturbations in Liu (2000, 2023), Liu et al. (2005, 2015), and Liu and Tang (2015), the effects of these main natural perturbations considered in satellite orbital motion, including the Earth's nonspherical gravitational perturbation, the additional perturbation of the coordinate system due to the precession and the nutation (especially the additional

perturbation showed in the potential of the J_2 term), the third-body gravitational perturbations from the Sun and the Moon, and the solar radiation pressure, fundamentally manifest as short-period terms for the orbital semi-major axis; while the atmospheric drag perturbation not only causes the long-term decay of the orbital semi-major axis but also possesses the short-period effect. However, the amplitudes of the short-period variations of the orbital semi-major axis for the Starlink satellites caused by the atmosphere drag perturbation are generally small, which will not have a substantial impact on orbital maneuver discriminations by setting an appropriate threshold. Over a short time (a 1-min interval set in the analytical retrieval below), the long-term decay of the orbital semi-major axis for the Starlink satellites caused by the atmosphere drag perturbation is generally negligible, and can be disregarded in the orbital maneuver discriminations. Of course, the corresponding results are more accurate taking the atmosphere drag effect into account. For simplifying the calculations, the atmosphere drag perturbation is not considered in the analytical retrieval of this paper. The feasibility of this simplification can be justified by the specific retrieval results obtained using the analytical and numerical methods below.

Therefore, based on the SpaceX-released ephemerides, and according to the analytical formulae for the short-period terms of orbital perturbations, the "mean orbital semi-major axis" a_m can be obtained by subtracting the short-period terms of these main natural perturbations except the atmospheric drag from the original orbital semi-major axis a_o (osculating orbital semi-major axis) of the Starlink satellites:

$$a_m = a_o - a_s^{J_l} - a_s^{J_{lm}} - a_s^{J_2\theta} - a_s^{Sun} - a_s^{Moon} - a_s^{SRP}, \tag{1}$$

where $a_s^{J_l}$, $a_s^{J_{lm}}$, $a_s^{J_2\theta}$, a_s^{Sun}, a_s^{Moon}, and a_s^{SRP} are the short-period terms of the zonal harmonics J_l ($l \geqslant 2$) of the Earth's nonspherical gravitational perturbation, the tesseral harmonics J_{lm} ($l \geqslant 2, m \geqslant 1$) of the Earth's nonspherical gravitational perturbation, the additional perturbation showed in the potential of the J_2 term, the third-body gravitational perturbation from the Sun, the third-body gravitational perturbation from the Moon, and the solar radiation pressure respectively; refer to Liu (2000, 2023), Liu et al. (2005, 2015), Liu and Tang (2015), and Zhang et al. (2018) for their specific expressions.

Because the effects of the main natural perturbations are excluded, low-thrust orbital maneuvers of the Starlink satellites can be clearly revealed with the "mean orbital semi-major axis". In the absence of orbital maneuvers, the variations for the time-series of the "mean orbital semi-major axis" remain relatively smooth. In contrast, when orbital maneuvers are implemented, the variations for the time-series of the "mean orbital semi-major axis" become extremely pronounced, see Figs. 1–4.

According to Starlink launch statistics and orbital state history of the Starlink satellites in the period of September 1st to 6th, 2024 (UTC) from the website[5], four representative satellites Starlink-4801, Starlink-11228, Starlink-32231 and Starlink-4682 (NORAD IDs 54825, 60398, 60439 and 53597) in the operational, parking, raising, and deorbit phases respectively are randomly selected for analysis. Using their ephemerides released by the SpaceX from September 1st to 5th, 2024 (UTC), the evolutions of the osculating orbital semi-major axis a_o and the "mean orbital semi-major axis" a_m over time t are presented in Figs. 1(a)-4(a) and 1(b)-4(b), respectively. Here, the Earth's nonspherical gravitational perturbation up to 20×20 order and degree, with the coefficients given by the Earth Gravitational Model 2008 (EGM2008) (Pavlis et al., 2012), is considered in the calculations of the short-period terms of the orbital semi-major axis. For the calculations of the short-period terms of the orbital semi-major axis caused by the third-body gravitational perturbations from the Sun and the Moon, the involved positions of the Sun and the Moon are derived from the JPL DE405 high-precision ephemeris[6]. For the calculations of the short-period terms of the orbital semi-major axis caused by the solar radiation pressure, the radiation pressure coefficient is set to 1.2, the area-to-mass ratios (A/m) of these four Starlink satellites are set as 0.033, 0.050, 0.050 and 0.033 m^2/kg, respectively, and the cylindrical Earth shadow model (Liu, 2000; Liu et al., 2005, 2015; Liu and Tang, 2015; Zhang et al., 2018) is adopted to determine whether the satellites move into the Earth's shadow. Note that the amplitudes of the short-period variations of the orbital semi-major axis caused by the solar radiation pressure are too small to have an observable effect on the retrieval results, when the A/m values are adjusted moderately. The obtained evolutions of the "mean orbital semi-major axis" a_m over time t in Figs. 1(b)-4(b) are similar to the related results in Liu et al. (2024b) and Liu et al. (2025).

The analysis of the time series for the short-period terms of the orbital semi-major axis, arising from the main natural perturbations, reveals distinct variation amplitudes across the different perturbations. Specifically, the variation amplitude is the largest for the Earth's zonal harmonics up to the 20th degree, with the magnitude reaching several kilometers. This is significantly greater than the variation amplitude for the Earth's tesseral harmonics up to the 20×20 degree and order, with the magnitude of hundreds of meters. Additionally, the additional perturbation showed in the potential of the J_2 term exhibit the even smaller variation amplitude, with the magnitude of 20–30 meters. In contrast, the third-body gravitational perturbations from the Sun and Moon and the solar radiation pressure show the smallest variation amplitudes with the magnitudes of just a few decimeters or as low as a single decimeter.

[5] https://planet4589.org/space/con/star/stats.html
[6] ftp://ssd.jpl.nasa.gov/pub/eph/

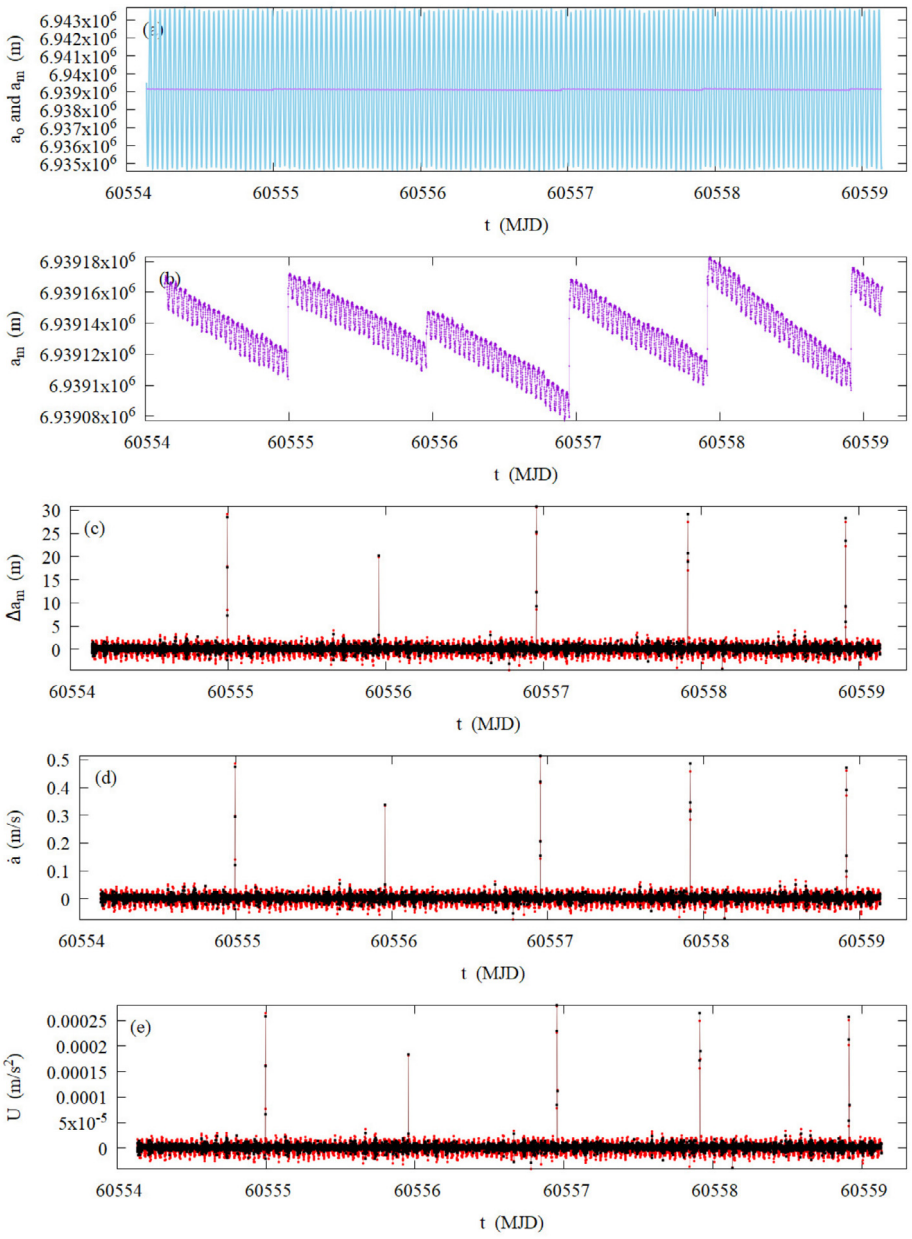


Fig. 1. For the satellite Starlink-4801 (NORAD ID 54825) in the operational phase: (a) the evolutions of the osculating orbital semi-major axis a_o and the "mean orbital semi-major axis" a_m over time t; (b) the evolution of the "mean orbital semi-major axis" a_m over time t; (c) the time series of the orbital semi-major axis variations per unit time (1 min) Δa_m; (d) the rate of change of the orbital semi-major axis $\dot{a}$; (e) the magnitude of the orbital maneuver acceleration $U = |\mathbf{a}_{th}|$. The sky-blue curve in the panel (a) corresponds to the evolution of the osculating orbital semi-major axis a_o over time t; the purple curves in the panels (a) and (b) correspond to the evolution of the "mean orbital semi-major axis" a_m over time t; the red curves in the panels (c)-(e) correspond to the results obtained by the analytical retrieval method, while the black curves correspond to the results obtained by the numerical retrieval method.

Therefore, although the specific dynamical model adopted in the SpaceX-released ephemerides is unknown, it can clearly reveal orbital maneuvers of the Starlink satellites by the time series of the "mean orbital semi-major axis" with the short-period effect of the Earth's gravitational field up to the 20×20 degree and order removed. The interferences of the short-period variations of the orbital semi-major axis caused by the third-body gravitational perturbations from the Sun and Moon, the solar radiation pressure and the atmospheric drag are negligible. These are the reasons why Liu et al. (2024b) can successfully detect and extract the fundamental strategies of orbital maneuvers of the Starlink satellites.

However, it is for this reason that the specific dynamical model adopted in the SpaceX-released ephemerides is unknown, the present analytical retrieval of orbital maneuver information considers not only the Earth's nonspherical gravitational perturbation up to the 20×20 degree

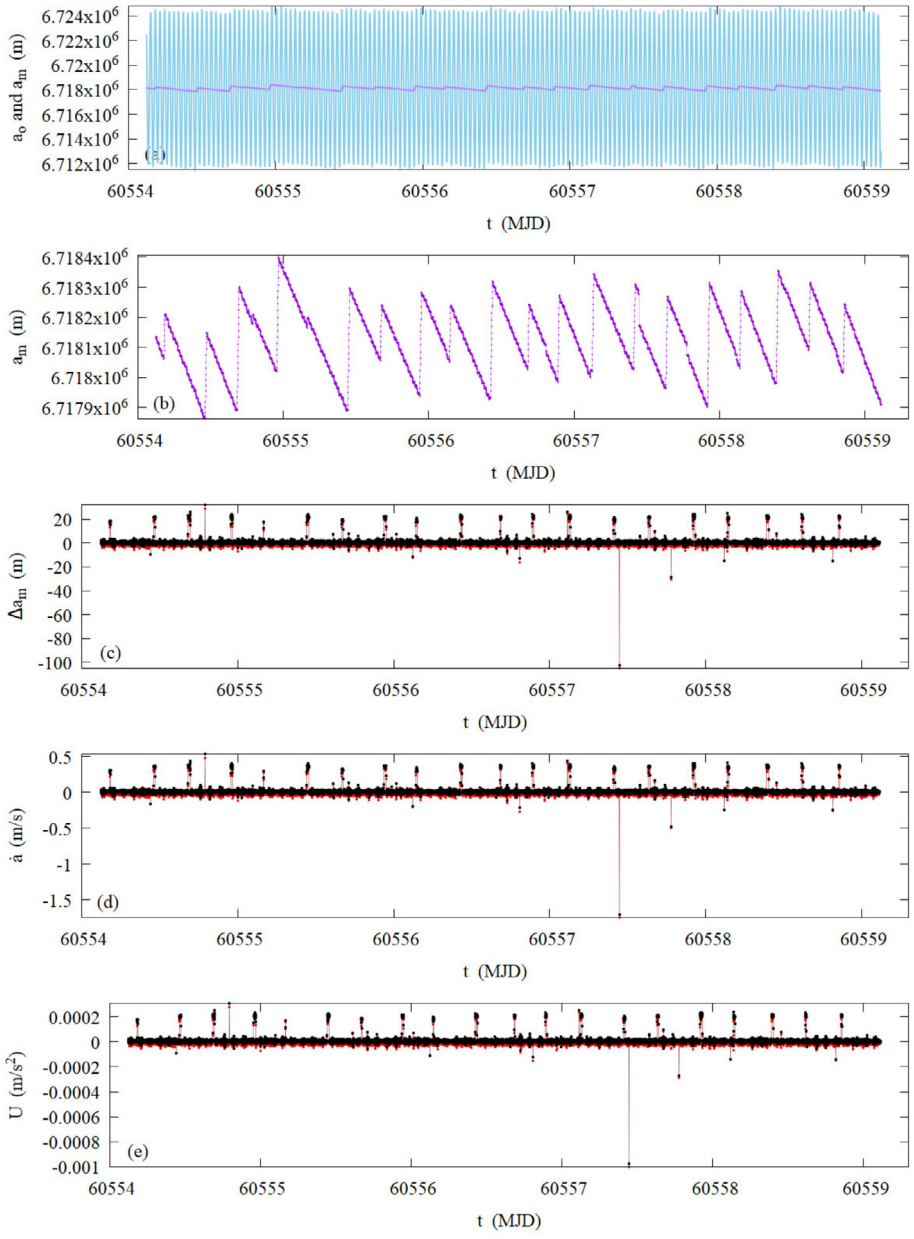


Fig. 2. For the satellite Starlink-11228 (NORAD ID 60398) in the parking phase: (a) the evolutions of the osculating orbital semi-major axis a_o and the "mean orbital semi-major axis" a_m over time t; (b) the evolution of the "mean orbital semi-major axis" a_m over time t; (c) the time series of the orbital semi-major axis variations per unit time (1 min) Δa_m; (d) the rate of change of the orbital semi-major axis $\dot{a}$; (e) the magnitude of the orbital maneuver acceleration $U = |\mathbf{a}_{th}|$. The curves with different colors have the same meaning as in Fig. 1.

and order, but also these important natural perturbations generally taken into account in satellite orbital dynamics, i.e., the additional perturbation showed in the potential of the J_2 term, the third-body gravitational perturbations from the Sun and Moon and the solar radiation pressure.

Using the analytical expressions for the short-period perturbation terms in satellite orbital dynamics, the time series of the "mean orbital semi-major axis" of the Starlink satellites $\{a_{m,i}\}$ $(i = 1, 2, \cdots)$ are derived as shown in Figs. 1(b)-4(b), and then the orbital maneuvers can be further detected and extracted according to their specific variations. As stated above, Liu et al. (2024b) performed piecewise linear fitting on the time series of the "mean orbital semi-major axis" $\{a_{m,i}\}$, and combined the corresponding linear fit slopes to detect and extract orbital maneuvers of the Starlink satellites. Essentially, it provides a more convenient and direct approach to retrieve the orbital maneuver information of the Starlink satellites through the variations of the "mean orbital semi-major axis" Δa_m per unit time Δt. Theoretically, the "mean orbital semi-major axis" is obtained by removing the main natural perturbation effects from the osculating orbital semi-major

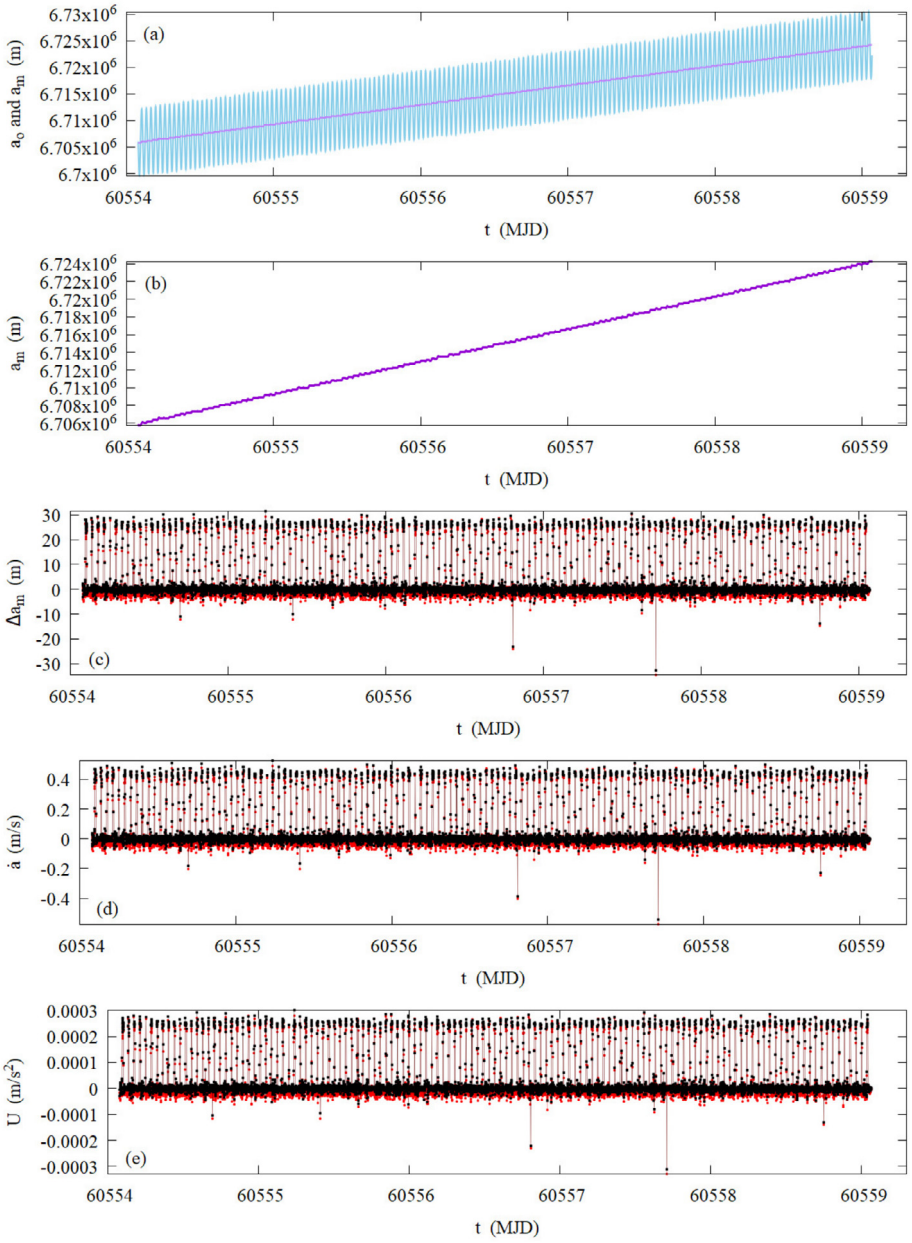


Fig. 3. For the satellite Starlink-32231 (NORAD ID 60439) in the orbit raising phase: (a) the evolutions of the osculating orbital semi-major axis a_o and the "mean orbital semi-major axis" a_m over time t; (b) the evolution of the "mean orbital semi-major axis" a_m over time t; (c) the time series of the orbital semi-major axis variations per unit time (1 min) Δa_m; (d) the rate of change of the orbital semi-major axis $\dot{a}$; (e) the magnitude of the orbital maneuver acceleration $U = |\mathbf{a}_{th}|$. The curves with different colors have the same meaning as in Fig. 1.

axis, and hence its variations per unit time are arising from the maneuver force and the atmospheric drag (which can be approximated as the maneuver force).

Thereby, we make an appropriate amendment to the orbital maneuver extraction method for the Starlink satellites based on the SpaceX-released ephemerides proposed by Liu et al. (2024b) as follows. From the time series of the "mean orbital semi-major axis" $\{a_{m,i}\}$, we directly obtain the time series of the orbital semi-major axis variations $\{\Delta a_{m,i} = a_{m,i} - a_{m,i-1}\}$ $(i = 2, 3, \cdots)$ per unit time $\Delta t = t_i - t_{i-1} = 1$ min caused by the maneuver force and the atmospheric drag (which can be approximated as the maneuver force). Then, setting a reasonable threshold for the variation of the "mean orbital semi-major axis", we perform detection and analysis for the time series $\{\Delta a_{m,i}\}$ to retrieve and extract the orbital maneuvers of the Starlink satellites. Here, there are two reasons for the choice of unit time Δt as 1 min. On the one hand, the Δa_m caused by the low-thrust orbital maneuvers of the Starlink satellites is sufficiently apparent within this timespan. On the other hand, the time step of the SpaceX-released ephemerides is 1 min, which ensures that the time series of the orbital semi-major axis variations caused by the maneuver force and the atmospheric drag (which can be approximated as

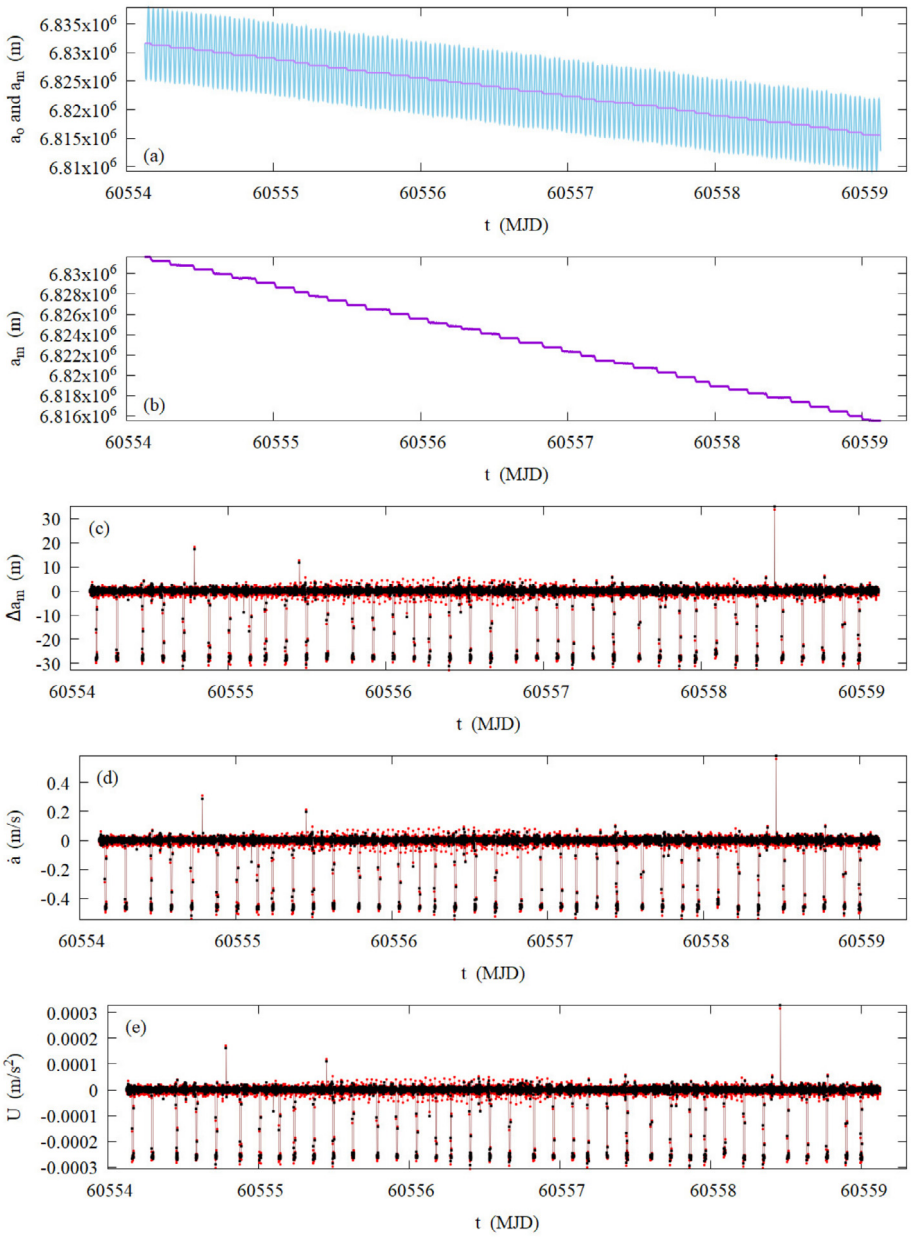


Fig. 4. For the satellite Starlink-4682 (NORAD ID 53597) in the deorbit phase: (a) the evolutions of the osculating orbital semi-major axis a_o and the "mean orbital semi-major axis" a_m over time t; (b) the evolution of the "mean orbital semi-major axis" a_m over time t; (c) the time series of the orbital semi-major axis variations per unit time (1 min) Δa_m; (d) the rate of change of the orbital semi-major axis $\dot{a}$; (e) the magnitude of the orbital maneuver acceleration $U = |\mathbf{a}_{th}|$. The curves with different colors have the same meaning as in Fig. 1.

the maneuver force), obtained by the analytical retrieval method, have equal time intervals. The orbital semi-major axis variations $\Delta a_{m,i}$ are thus approximately the cumulative effect of the maneuver force over the equal time intervals $\Delta t = t_i - t_{i-1} = 1$ min, which facilitates the analytical retrieval to reveal the fundamental information of orbital maneuvers of the Starlink satellites.

The time series of the orbital semi-major axis variations $\{\Delta a_{m,i}\}$ per 1 min, caused by the maneuver force and the atmospheric drag (which can be approximated as the maneuver force), for the four satellites Starlink-4801, Starlink-11228, Starlink-32231 and Starlink-4682 in the operational, parking, orbit raising and deorbit phases, respectively, are presented in Figs. 1(c)-4(c) with the red curves. The orbital maneuvers of the Starlink satellites are intuitively and clearly revealed. This is notable that such orbital maneuvers are generally continuous low-thrust operations, as depicted in the locally magnified view of the time series of the orbital semi-major axis variations per 1 min for the satellite Starlink-32231 in the orbit raising phase in Fig. 5 with the red curves, taking place over an extended period of time rather than impulsive or quasi-impulsive maneuvers. For the Starlink satellites in the operational and parking phases, due to the long-term

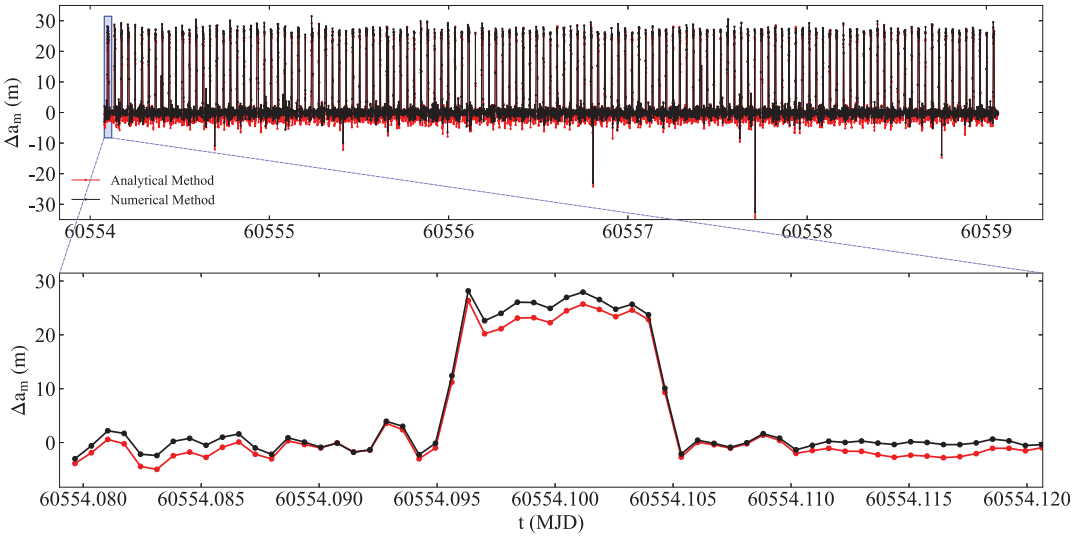


Fig. 5. The locally magnified view of the time series of the orbital semi-major axis variations per unit time (1 min) Δa_m for the satellite Starlink-32231 (NORAD ID 60439) in the orbit raising phase. The red curves correspond to the results obtained by the analytical retrieval method, while the black curves correspond to the results obtained by the numerical retrieval method.

decay of the orbital semi-major axis, the orbital maneuvers are implemented to raise the orbital semi-major axis at regular intervals. Because its orbital altitude is lower for the Starlink satellite in the parking phase, the effect of the atmospheric drag is stronger, and the orbital maneuvers are more frequent than the Starlink satellite in the operational phase. There are also "abnormal" orbital maneuvers to reduce the orbital semi-major axis in Fig. 2(c). For the Starlink satellites in the orbit raising and deorbit phases, the orbital maneuvers are implemented more frequently to respectively raise and reduce the orbital semi-major axis as soon as possible, but the "abnormal" orbital maneuvers also exist to reduce and raise the orbital semi-major axes in Figs. 3(c) and 4(c), respectively. The specific reasons behind these observed "abnormal" orbital maneuvers are unknown, potentially stemming from collision-avoidance mechanisms that deserve further research in the future.

According to the orbital semi-major axis variation Δa_m per unit time ($\Delta t = 1$ min), the rate of change of the orbital semi-major axis $\dot{a}$ can be determined via $\dot{a} = \frac{\mathrm{d}a}{\mathrm{d}t} \approx \frac{\Delta a_m}{\Delta t}$. The derived rate of change of the orbital semi-major axis $\dot{a}$, caused by the maneuver force and the atmospheric drag (which can be approximated as the maneuver force), represents an average value over the time interval $\Delta t = 1$ min. This rate of change of the orbital semi-major axis $\dot{a}$ is assumed to closely approximate the instantaneous rate of change at each epoch within that time interval. Then, it is assumed that the orbital maneuvers only act in the tangential direction of motion of the Starlink satellites, and the value of the maneuver acceleration $|\mathbf{a}_{th}| = U$ can be further calculated using the following formula (Liu, 2000; Liu et al., 2015; Liu, 2023):

$$\dot{a} = \frac{\mathrm{d}a}{\mathrm{d}t} = \frac{2}{n\sqrt{1-e^2}}\left(1 + 2e\cos f + e^2\right)^{1/2} U, \tag{2}$$

where a, e, and f are the satellite's orbital semi-major axis, eccentricity and true anomaly, respectively; $n = \sqrt{\mu} a^{-3/2}$ is the satellite's mean angular speed, and μ is the Earth's gravitational constant. In the process of computing the time series of the orbital semi-major axis variations $\{\Delta a_{m,i}\}$ per unit time $\Delta t = 1$ min, caused by the maneuver force and the atmospheric drag (which can be approximated as the maneuver force), by the analytical retrieval method, the corresponding rate of change of the orbital semi-major axis $\dot{a}$ and the value of the maneuver acceleration $|\mathbf{a}_{th}| = U$ are also calculated simultaneously, see the red curves in Figs. 1(d)-4(d) and 1(e)-4(e).

Similar to Liu et al. (2024b), setting the threshold for the variation of the "mean orbital semi-major axis" as 10 m, the orbital maneuvers are easily identified whenever the absolute variation exceeds this threshold, i.e., $|\Delta a_{m,i}| \geqslant 10$ m. In examining the time series of the orbital semi-major axis variations $\{\Delta a_{m,i}\}$ per unit time $\Delta t = 1$ min, the first epoch where the absolute variation $|\Delta a_{m,i}|$ exceeds the threshold is identified as the start time of the maneuver, labeled as t_{start}. The last epoch where the absolute variation $|\Delta a_{m,i}|$ remains above the threshold is marked as the end time of the maneuver, labeled as t_{end}. The duration of the maneuver ΔT is therefore $\Delta T = t_{end} - t_{start}$. During the period of ΔT, all the absolute variations $|\Delta a_{m,i}|$ consistently exceed the threshold of 10 m. Further, the start and end times, the duration of the maneuver ΔT, the total variation of the orbital semi-major axis Δa, the average rate of change of the orbital semi-major axis $\bar{\dot{a}}$ and the average maneuver acceleration $|\bar{\mathbf{a}}_{th}|$ for each maneuver can be acquired. Here, the $\bar{\dot{a}}$ and $|\bar{\mathbf{a}}_{th}|$ for each maneuver are the average values of the $\dot{a}$ and $|\mathbf{a}_{th}|$ in the ΔT, i.e.,

$$\bar{\dot{a}} = \frac{\sum_{\Delta T} \dot{a}}{\Delta T}, \quad |\bar{\mathbf{a}}_{th}| = \frac{\sum_{\Delta T} |\mathbf{a}_{th}|}{\Delta T} = \frac{\sum_{\Delta T} U}{\Delta T}. \tag{3}$$

The detailed orbital maneuvers for the satellite Starlink-4801 from September 1st to 5th, 2024 (UTC) are presented in Table 1. There are also the similar details for the satellites Starlink-11228, Starlink-32231 and Starlink-4682, but they are relatively lengthy and presented in Tables A.1, A.2, and A.3 in Appendix A.

From Figs. 1–4, it is evident that the orbital maneuvers of these four Starlink satellites, in the operational, parking, orbit raising, and deorbit phases, respectively, exhibit distinct fundamental information, especially the average time interval between the orbital maneuvers $\Delta\overline{S}$, the average variation of the orbital semi-major axis $\Delta\bar{a}$, the average duration $\Delta\overline{T}$, the average rate of change of the orbital semi-major axis $\dot{\bar{a}}$ and the average maneuver acceleration $|\bar{\mathbf{a}}_{th}|$ of the maneuvers, and their standard deviations ($\sigma_{\Delta\bar{S}}$, $\sigma_{\Delta\bar{a}}$, $\sigma_{\Delta\bar{T}}$, $\sigma_{\dot{\bar{a}}}$, and $\sigma_{|\bar{\mathbf{a}}_{th}|}$), see Table 2. Note that the "abnormal" descending maneuvers for the satellites Starlink-11228 and Starlink-32231 in the parking and orbit raising phases and the "abnormal" ascending maneuvers for the satellite Starlink-4682 in the deorbit phase are excluded in the statistics of the fundamental maneuver information; and the three-sigma rule is further activated to eliminate outliers and derive the final values in the statistics of the $\Delta\overline{S}$ and $\sigma_{\Delta\overline{S}}$ for the satellites Starlink-32231 and Starlink-4682. These fundamental information of orbital maneuvers is beneficial for the orbit determination and prediction of the Starlink satellites with some valuable references. Specifically, the $\Delta\overline{S}$ for the four satellites Starlink-4801, Starlink-11228, Starlink-32231 and Starlink-4682 in the operational, parking, orbit raising and deorbit phases are about 0.978, 0.204, 0.032 and 0.115 day from the analytical retrieval, respectively. Moreover, the $\Delta\bar{a}$, $\Delta\overline{T}$, $\dot{\bar{a}}$, and $|\bar{\mathbf{a}}_{th}|$ are also different for the different satellites.

In summary, the core essence of the analytical method for the retrieval of the orbital maneuver information of the Starlink satellites based on the SpaceX-released ephemerides is as follows. The time series of the orbital semi-major axis variations per unit time (1 min), caused by the maneuver force and the atmospheric drag (which can be approximated as the maneuver force), are analytically retrieved through the calculations of the short-period perturbation terms in satellite orbital dynamics with the analytical formulae, thereby exploring the orbital maneuver information of the Starlink satellites. The amendments to the orbital maneuver extraction method proposed by Liu et al. (2024b) are primarily summarized as follows. The analytical retrieval method in this paper further takes into account the short-period terms of the additional perturbation showed in the potential of the J_2 term, the third-body gravitational perturbations from the Sun and Moon and the solar radiation pressure. Importantly, the analytical retrieval method in this paper particularly employs the variations of the "mean orbital semi-major axis" per unit time Δt to replace the slopes of the piecewise linear fittings for the maneuver discriminations. The advantage of this approach is that it is more convenient and straightforward

Table 1
The detailed orbital maneuvers for the satellite Starlink-4801 (NORAD ID 54825) in the operational phase retrieved from the ephemerides released by the SpaceX from September 1st to 5th, 2024 (UTC).

Analytical retrieval						Numerical retrieval					
Start time	End time	ΔT (min)	Δa (m)	$\dot{\bar{a}}$ (m/s)	$\lvert\bar{\mathbf{a}}_{th}\rvert$ (m/s^2)	Start time	End time	ΔT (min)	Δa (m)	$\dot{\bar{a}}$ (m/s)	$\lvert\bar{\mathbf{a}}_{th}\rvert$ (m/s^2)
2024–09-01 23:50:42.00	2024–09-01 23:52:42.00	2	46.884	0.391	2.13E-04	2024–09-01 23:50:42.00	2024–09-01 23:52:42.00	2	46.086	0.384	2.09E-04
2024–09-02 22:56:42.00	2024–09-02 22:57:42.00	1	19.974	0.333	1.82E-04	2024–09-02 22:56:42.00	2024–09-02 22:57:42.00	1	20.272	0.338	1.84E-04
2024–09-03 22:51:42.00	2024–09-03 22:54:42.00	3	67.966	0.378	2.06E-04	2024–09-03 22:51:42.00	2024–09-03 22:54:42.00	3	68.414	0.380	2.07E-04
2024–09-04 21:52:42.00	2024–09-04 21:55:42.00	3	63.750	0.354	1.93E-04	2024–09-04 21:52:42.00	2024–09-04 21:55:42.00	3	68.841	0.382	2.09E-04
2024–09-05 21:55:42.00	2024–09-05 21:57:42.00	2	49.798	0.415	2.26E-04	2024–09-05 21:55:42.00	2024–09-05 21:57:42.00	2	51.710	0.431	2.35E-04

Table 2
Retrieval results of the fundamental information of orbital maneuvers for the Starlink satellites in the different phases.

NORAD ID	Method	$\Delta\overline{S}$ (Day)	$\sigma_{\Delta\overline{S}}$ (Day)	$\Delta\bar{a}$ (m)	$\sigma_{\Delta\bar{a}}$ (m)	$\Delta\overline{T}$ (min)	$\sigma_{\Delta\overline{T}}$ (min)	$\bar{\tilde{a}}$ (m/s)	$\sigma_{\bar{\tilde{a}}}$ (m/s)	$\lvert\bar{\mathbf{a}}_{th}\rvert$ (m/s^2)	$\sigma_{\lvert\bar{\mathbf{a}}_{th}\rvert}$ (m/s^2)
54825	Analytical	0.978	0.023	49.674	18.859	2.200	0.837	0.374	0.032	0.204E-03	0.174E-04
	Numerical	0.978	0.023	51.065	19.940	2.200	0.837	0.383	0.033	0.209E-03	0.180E-04
60398	Analytical	0.204	0.077	246.020	122.866	12.609	6.013	0.329	0.048	0.189E-03	0.274E-04
	Numerical	0.204	0.077	264.571	130.874	12.696	6.018	0.352	0.052	0.202E-03	0.302E-04
60439	Analytical	0.032	0.001	186.049	63.540	7.889	2.634	0.390	0.030	0.224E-03	0.173E-04
	Numerical	0.032	0.001	195.615	66.107	8.000	2.626	0.404	0.028	0.232E-03	0.161E-04
53597	Analytical	0.115	0.012	−387.977	50.385	14.447	1.884	−0.448	0.010	−0.251E-03	0.576E-05
	Numerical	0.111	0.019	−373.917	76.806	14.051	2.790	−0.437	0.044	−0.245E-03	0.246E-04

for the maneuver discriminations, avoiding lots of calculations of piecewise linear fittings and reducing dependence on the number of data points in the maneuver segments. Nevertheless, there is a disadvantage that the discrimination of the start and end times of the orbital maneuvers depends on the chosen unit time interval (i.e., the temporal resolution). In this paper, the temporal resolution for the orbital maneuver discriminations is set to 1 min, whereas the method proposed by Liu et al. (2024b) is not subject to this constraint, allowing for potentially higher temporal resolution in the orbital maneuver discriminations. However, the temporal resolution of 1 min employed in this paper for the orbital maneuver discriminations proves to be sufficiently effective. This is primarily because the resolution is short enough to accurately determine the start and end times of the maneuvers, while being long enough to ensure that the signatures caused by these maneuvers stand out above the noise.

2.1.2. Numerical method

The core essence of the numerical method for the retrieval of the orbital maneuver information of the Starlink satellites based on the SpaceX-released ephemerides is as follows. The time series of the orbital semi-major axis variations per unit time (it is also set to 1 min, and the reasons for this choice are as stated above), caused solely by the maneuver force, are numerically retrieved through orbit propagation calculations, thereby exploring the orbital maneuver information of the Starlink satellites.

Taking the position and velocity vectors $\left(\vec{r}_1, \dot{\vec{r}}_1\right)$, $\left(\vec{r}_2, \dot{\vec{r}}_2\right)$, $\left(\vec{r}_3, \dot{\vec{r}}_3\right)$, $\cdots$ at the epochs t_1, t_2, t_3, $\cdots$ of the Starlink satellites in the SpaceX-released ephemerides as the initial values for orbit propagations successively, the satellite orbital states (position and velocity vectors) $\left(\vec{r}'_2, \dot{\vec{r}}'_2\right)$, $\left(\vec{r}'_3, \dot{\vec{r}}'_3\right)$, $\left(\vec{r}'_4, \dot{\vec{r}}'_4\right)$, $\cdots$ at the epochs t_2, t_3, t_4, $\cdots$ are derived by high-precision numerical integrations with the main natural perturbations. Namely, the time intervals for these numerical integrations are $\Delta t = (t_2 - t_1)$, $(t_3 - t_2)$, $(t_4 - t_3)$, $\cdots$, and the Δt is set as 1 min. And then, the position and velocity vectors of the satellites are converted to the Keplerian orbital elements. According to the differences between the actual orbital elements $\boldsymbol{K}_i$ and the calculated orbital elements $\boldsymbol{K}'_i$ by the numerical propagations at the corresponding epochs, i.e., deducting the calculated values of $\boldsymbol{K}'_2$, $\boldsymbol{K}'_3$, $\boldsymbol{K}'_4$, $\cdots$, given by the high-precision numerical integrations with the main natural perturbations, from the converted orbital elements $\boldsymbol{K}_2$, $\boldsymbol{K}_3$, $\boldsymbol{K}_4$, $\cdots$ containing the maneuver effects in the SpaceX-released ephemerides at the epochs t_2, t_3, t_4, $\cdots$, the time series of the variations of the orbital elements within the corresponding time intervals $\Delta t = (t_2 - t_1)$, $(t_3 - t_2)$, $(t_4 - t_3)$, $\cdots = 1$ min, caused solely by the maneuver force, are finally acquired:

$$\left\{\Delta\boldsymbol{K}_{m,i} = \boldsymbol{K}_i - \boldsymbol{K}'_i\right\}, \quad i = 2, 3, 4 \cdots. \tag{4}$$

Here, we focus on the time series of the orbital semi-major axis variations $\{\Delta a_{m,i}\}$ caused solely by the maneuver force. Evidently, the orbital semi-major axis variations are the cumulative effects of the maneuver force over the time intervals $\Delta t = t_i - t_{i-1} = 1$ min.

Besides these main natural perturbations considered in the above analytical retrieval method, i.e., the Earth's non-spherical gravitational perturbation up to the 20×20 degree and order, the third-body gravitational perturbations from the Sun and Moon and the solar radiation pressure, the atmospheric drag perturbation is also taken into account in the high-precision numerical integrations with the Runge–Kutta-Fehlberg 7(8) method. Refer to Liu (2000, 2023), Liu et al. (2005, 2015), Liu and Tang (2015) and Montenbruck and Gill (2000) for the specific mathematical formulations of these perturbation models used in the numerical integrations. The corresponding parameters, including the Earth's gravitational field coefficients, the ephemeris of calculating the positions of the Sun and the Moon, the radiation pressure coefficient and the A/m values involved in the solar radiation pressure are the same as the ones adopted in the analytical retrieval. It is worth noting that the additional perturbation showed in the potential of the J_2 term, considered in the above analytical retrieval, is no longer appearing independently in the numerical integrations in the geocentric celestial reference system, but included in the calculation of the Earth's non-spherical gravitational perturbation (Liu et al., 2005). For the atmospheric drag perturbation, the NRLMSISE-00 empirical model of the atmosphere (Picone et al., 2002) is adopted to acquire the involved

values of atmospheric densities, and the A/m values of the four satellites are set as 0.033, 0.010, 0.008 and 0.010 m^2/kg, respectively. The atmospheric drag coefficient C_D is not provided in the SpaceX-released ephemerides, and hence it is set to a basic value of 2.2 for the calculations. The associated modeling uncertainties are balanced by adjusting the A/m values involved in the atmospheric drag perturbation. Actually, the equivalent A/m values involved in the atmospheric drag perturbation are determined empirically, according to the fact that the orbital semi-major axis variations approach to zero in the absence of orbital maneuvers, i.e., $\{\Delta a_{m,i} \approx 0\}$in the non-maneuver arcs. In other words, the equivalent $C_D(A/m)$ information of the Starlink satellites involved in the atmospheric drag perturbation can be also estimated effectively by the numerical retrieval of the orbital maneuvers.

Through the numerical retrieval, the time series of the orbital semi-major axis variations $\{\Delta a_{m,i}\}$ per 1 min caused solely by the maneuver force, the corresponding rate of change of the orbital semi-major axis $\dot{a}$ (it is also assumed that the average rate of change of the orbital semi-major axis over the time interval $\Delta t = 1$ min, caused by the maneuver force, closely approximates the instantaneous rate of change at each epoch within that time interval), and the value of the maneuver acceleration $|\mathbf{a}_{th}| = U$ (it is also assumed that the orbital maneuvers only act in the tangential direction of motion), for the four satellites Starlink-4801, Starlink-11228, Starlink-32231 and Starlink-4682 in the operational, parking, orbit raising and deorbit phases, respectively, are presented in Figs. 1(c)-4(c), 1(d)-4(d), and 1(e)-4(e) with the black curves. Furthermore, the locally magnified view of the time series of the orbital semi-major axis variations per 1 min for the satellite Starlink-32231 in the orbit raising phase is also presented in Fig. 5 with the black curves. It can be seen that the time series of the orbital semi-major axis variations, the rate of change of the orbital semi-major axis and the value of the maneuver acceleration retrieved by the analytical and numerical methods show good agreements. Further taking the atmospheric drag perturbation into account in the numerical retrieval, it can better avoid its interference on the orbital maneuver discriminations.

Similarly, setting the threshold for the variation of the orbital semi-major axis as 10 m, the orbital maneuvers are identified when $|\Delta a_{m,i}| \geqslant 10$ m, and the details, including the start and end times, the duration of the maneuver ΔT, the total variation of the orbital semi-major axis Δa, the average rate of change of the orbital semi-major axis $\bar{\dot{a}}$ and the average maneuver acceleration $|\bar{\mathbf{a}}_{th}|$ for each maneuver, can be also acquired by the numerical retrieval method. The detailed orbital maneuvers obtained by the numerical retrieval for the satellite Starlink-4801 from September 1st to 5th, 2024 (UTC) are also presented in Table 1. There are also the similar details for the satellites Starlink-11228, Starlink-32231 and Starlink-4682, as shown in Tables A.1, A.2, and A.3 in Appendix A. The detailed orbital maneuvers obtained by the numerical retrieval coincide with the results from the analytical retrieval on the whole. Correspondingly, for these four Starlink satellites in the operational, parking, orbit raising and deorbit phases, respectively, the fundamental information of orbital maneuvers revealed by the analytical and numerical retrieval methods are consistent on the whole (see Table 2). The differences of the retrieval results between the analytical and numerical methods mainly arise from the atmospheric drag effect, especially the Δa, $\Delta\bar{a}$, $\bar{\dot{a}}$, and $|\bar{\mathbf{a}}_{th}|$ in Tables 1 and 2. Moreover, the magnitudes of the atmospheric drag for the different satellites can be derived by comparing the values of $|\bar{\mathbf{a}}_{th}|$ from the analytical retrieval and the numerical retrieval.

2.2. Computational procedure

Computational procedure of the retrieval of orbital maneuver information of the Starlink satellites from the SpaceX-released ephemerides is briefly summarized as follows.

(1) Data integrity check and ephemeris splicing. Integrity checks on file packages of the SpaceX-released ephemerides, downloaded continuously over a period of time, are performed to ensure no missing data. For each Starlink satellite to be discussed, according to the principle of retaining the latest released data, ephemeris information covering non-overlapping time periods are extracted from multiple ephemeris data files, and spliced to form a single ephemeris data file with continuous overall coverage. In the specific operation process, for the ephemeris files released at adjacent times, the ephemeris start epoch T_b is get from the satellite ephemeris file released at adjacent next time, then the ephemeris data prior to T_b from the satellite ephemeris file released at adjacent previous time are extracted, and sequentially spliced to form the final ephemeris data file used for analysis.

(2) Retrieval of the orbital semi-major axis variations caused by the maneuver force. For the Starlink satellite to be discussed, based on its spliced ephemeris data and using the basic principles described in Section 2.1, the time series of the orbital semi-major axis variations $\{\Delta a_{m,i}\}$, caused by the maneuver force (or the approximated maneuver force), are retrieved by the analytical or numerical method, as well as the corresponding rate of change of the orbital semi-major axis $\dot{a}$ and the value of the maneuver acceleration $|\mathbf{a}_{th}| = U$.

(3) Orbital maneuver discrimination. Setting the threshold for the variation of the orbital semi-major axis (it is set to 10 m in this paper), orbital maneuver discriminations are performed on the basis of the retrieved time series of the orbital semi-major axis variations $\{\Delta a_{m,i}\}$. When the $\Delta a_{m,i}$ exceeds the set threshold, the orbital maneuvers are identified, and the associated details can be obtained, including the start and end times, the duration ΔT, the total variation of the orbital semi-major axis Δa, the average rate

of change of the orbital semi-major axis $\dot{\bar{a}}$ and the average maneuver acceleration $|\bar{\mathbf{a}}_{th}|$ for each maneuver.

(4) Acquirement of orbital maneuver information by statistical analysis. Similar to Liu et al. (2024b), based on the obtained orbital maneuvers of the Starlink satellites, further statistical analysis is performed, and the fundamental information of orbital maneuvers is finally acquired, especially the average time interval between the orbital maneuvers $\Delta\overline{S}$, the average variation of the orbital semi-major axis $\Delta\bar{a}$, the average duration $\Delta\overline{T}$, the average rate of change of the orbital semi-major axis $\dot{\bar{a}}$ and the average maneuver acceleration $|\bar{\mathbf{a}}_{th}|$ of the maneuvers, and their standard deviations ($\sigma_{\Delta\bar{S}}$, $\sigma_{\Delta\bar{a}}$, $\sigma_{\Delta\bar{T}}$, $\sigma_{\dot{\bar{a}}}$, and $\sigma_{|\bar{\mathbf{a}}_{th}|}$). Here, the statistical analysis of the orbital maneuver information can be performed not only for a single Starlink satellite to be discussed but also for a batch of Starlink satellites in the same phases with similar orbital states (e.g., Starlink satellites from the same launch batch).

3. Retrieval results of orbital maneuver information for more satellites

To gain a deeper understanding of orbital maneuvers for the same launch batches and their fundamental information

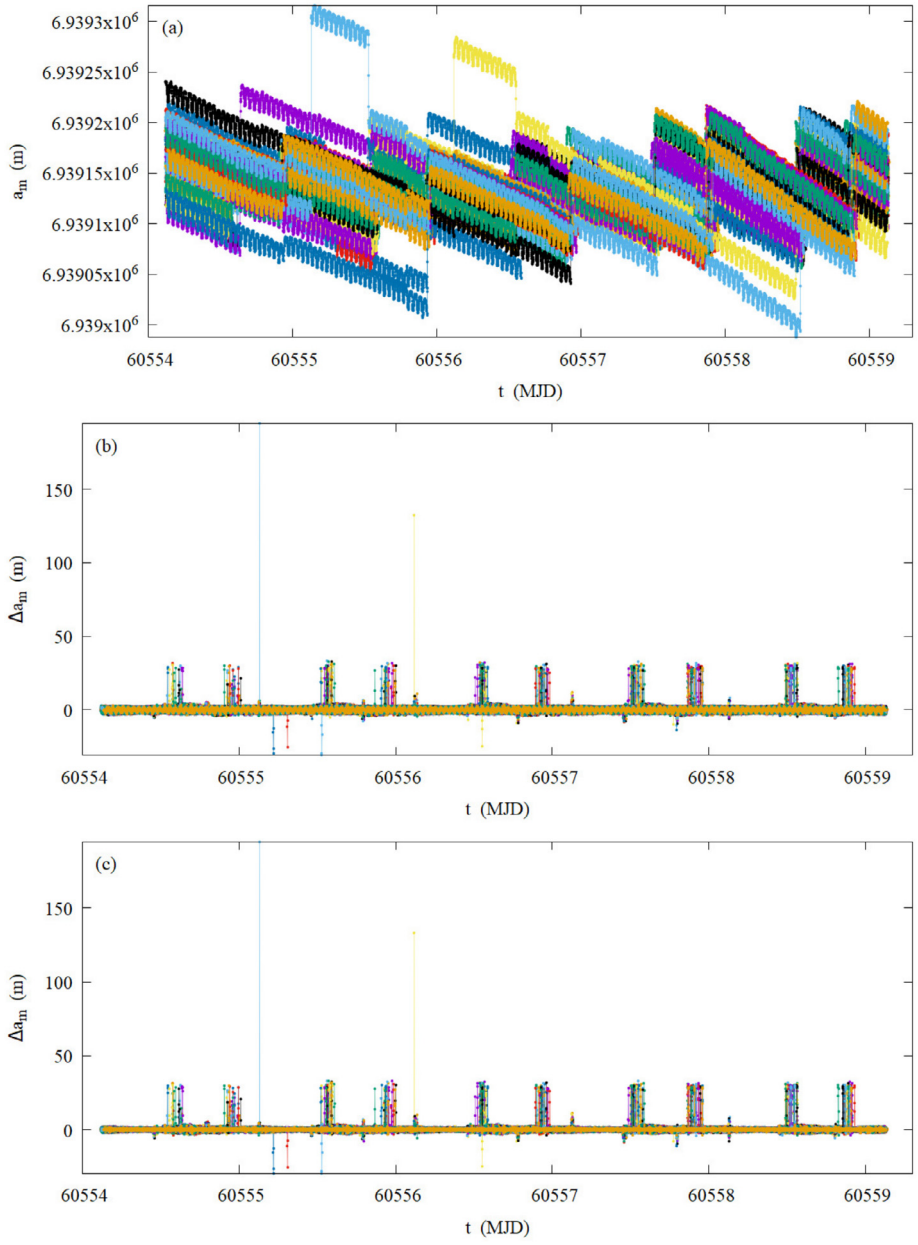


Fig. 6. For 52 satellites from Starlink Group 5–1 in the operational phase: (a) the evolution of the "mean orbital semi-major axis" a_m over time t retrieved by the analytical method; (b) the time series of the orbital semi-major axis variations over 1 min Δa_m caused by the maneuver force and the atmospheric drag (which can be approximated as the maneuver force), retrieved by the analytical method; (c) the time series of the orbital semi-major axis variations over 1 min Δa_m caused solely by the maneuver force, retrieved by the numerical method. The curves with different colors correspond to the different satellites.

across different phases, more satellites are randomly selected for further analysis, based on Starlink launch statistics and orbital state history accessed from the website[5]. From September 1st to 6th, 2024 (UTC), 52 satellites (NORAD IDs 54820–54828, 54830–54838, and 54840–54873) from Starlink Group 5–1 (Launch 69, 2022–177) were in the operational phase; 5 satellites from Starlink Group 8–3 (Launch 187, 2024–141), i.e., Starlink-11228, Starlink-11182, Starlink-111234, Starlink-111232 and Starlink-11139 (NORAD IDs 60398–60402), were in the parking phase; 23 satellites (NORAD IDs 60425–60447) from Starlink Group 10–7 (Launch 188, 2024–144) were in the orbit raising phase, but a satellite Starlink-32249 (NORAD ID 60444) lacks one released ephemeris file, so the ephemeris data of the remaining 22 satellites from this batch are used for the retrieval below; Starlink-3921, Starlink-4682, Starlink-5380, Starlink-2108 and Starlink-2132 (NORAD IDs 52613, 53597, 54829, 47387 and 47729) were in the deorbit phase. In the following, based on the corresponding spliced SpaceX-released ephemerides, the specific orbital maneuver information of these satellites are derived by the analytical and numerical retrieval methods.

Through analytical retrieval, the evolutions of the "mean orbital semi-major axis" a_m over time t and the time series of the orbital semi-major axis variations per unit time (1 min) Δa_m for these satellites are presented in Figs. 6(a)-9(a)

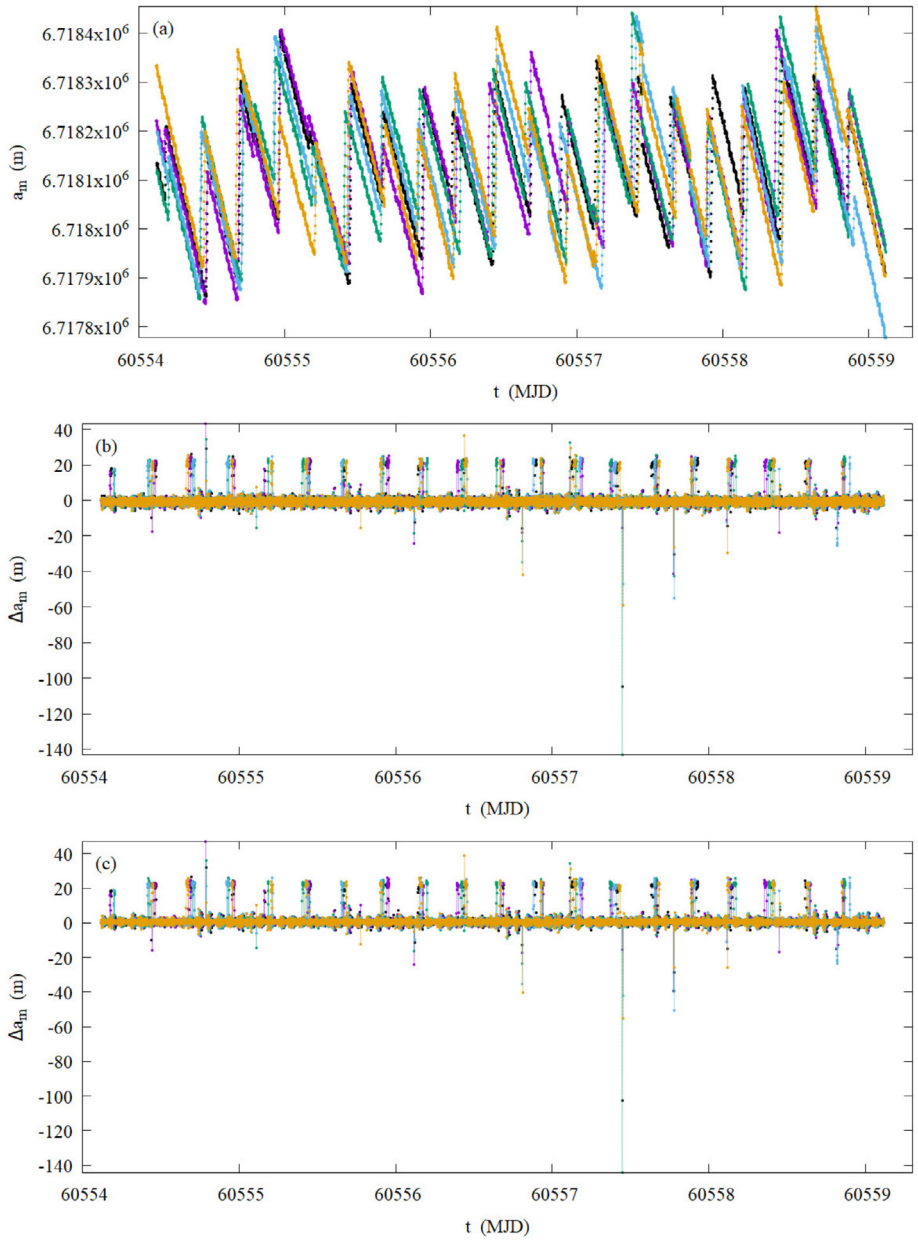


Fig. 7. For 5 satellites from Starlink Group 8–3 in the parking phase: (a) the evolution of the "mean orbital semi-major axis" a_m over time t retrieved by the analytical method; (b) the time series of the orbital semi-major axis variations over 1 min Δa_m caused by the maneuver force and the atmospheric drag (which can be approximated as the maneuver force), retrieved by the analytical method; (c) the time series of the orbital semi-major axis variations over 1 min Δa_m caused solely by the maneuver force, retrieved by the numerical method. The curves with different colors correspond to the different satellites.

and 6(b)-9(b), respectively. Through numerical retrieval, the corresponding time series of the orbital semi-major axis variations per unit time (1 min) Δa_m for these satellites are presented in Figs. 6(c)-9(c), respectively. From Figs. 6–9, it is seen that the orbital maneuvers show relatively obvious characteristics, particularly in terms of the time intervals between the orbital maneuvers; the maximum variations of the orbital semi-major axis over 1 min in each orbital maneuver are equivalent roughly, with eliminating the outliers. In addition, for the Starlink satellites from the same batches in the operational, parking and orbit raising phases, respectively, they perform orbital maneuvers at different epochs. The average time intervals between the orbital maneuvers $\Delta\overline{S}$ for the 52 satellites from Starlink Group 5–1 in the operational phase are approximately equal. The orbital maneuvers of these satellites exhibit the notable concentration phenomenon in terms of their epochs, partitioned into two groups. From the perspective

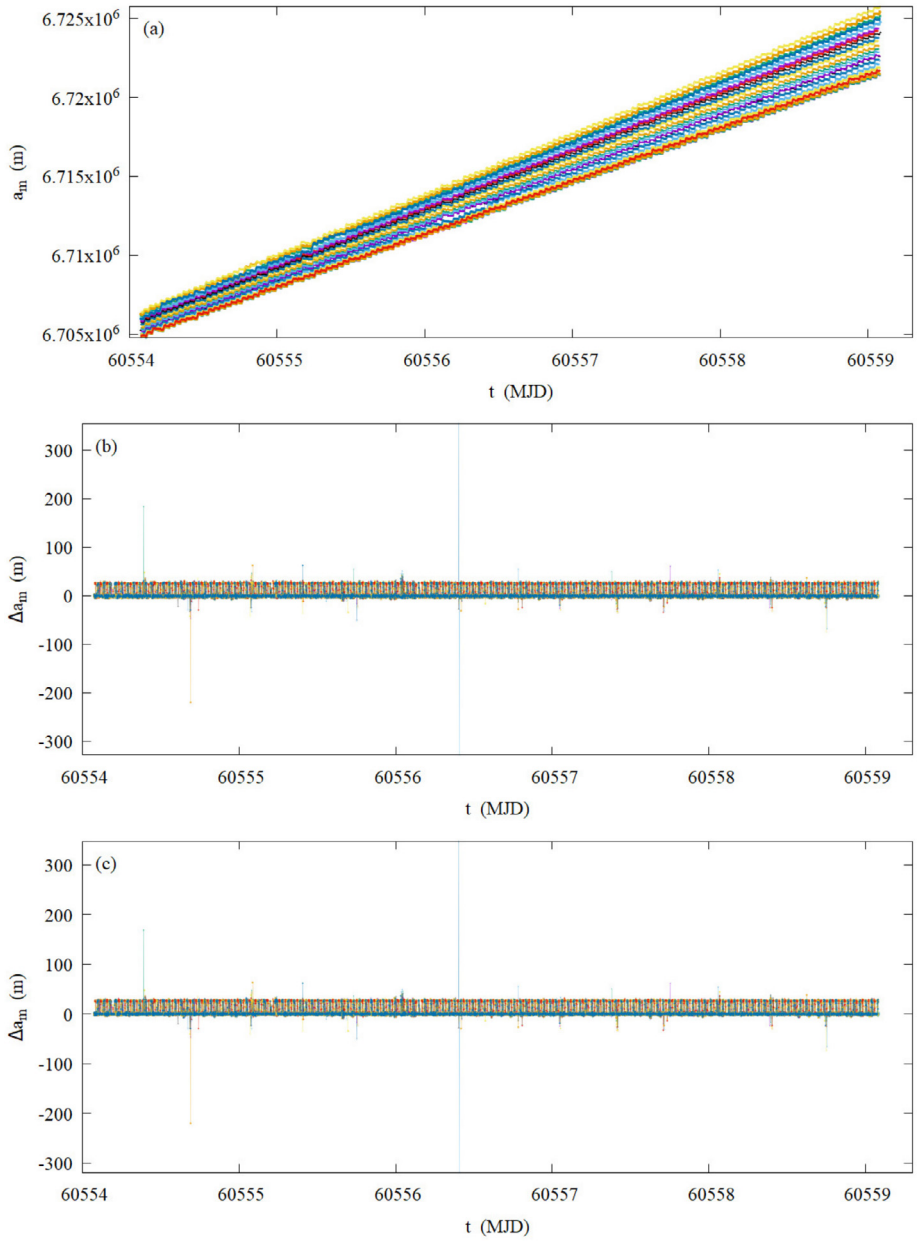


Fig. 8. For 22 satellites from Starlink Group 10–7 in the orbit raising phase: (a) the evolution of the "mean orbital semi-major axis" a_m over time t retrieved by the analytical method; (b) the time series of the orbital semi-major axis variations over 1 min Δa_m caused by the maneuver force and the atmospheric drag (which can be approximated as the maneuver force), retrieved by the analytical method; (c) the time series of the orbital semi-major axis variations over 1 min Δa_m caused solely by the maneuver force, retrieved by the numerical method. The curves with different colors correspond to the different satellites.

of orbital maneuver epochs, the 52 satellites from Starlink Group 5–1 in the operational phase can be generally divided into two relatively concentrated groups.

Furthermore, according to the principle that the minority is subordinate to the major, the "abnormal" orbital maneuvers for the satellites from Starlink Group 5–1 in the operational phase, including all data for the satellites Starlink-5407 and Starlink-5359 (NORAD IDs 54858 and 54864), individual data for the satellites Starlink-5106, Starlink-5369 and Starlink- 4659 (NORAD IDs 54823, 54837 and 54841) and the related orbit descending maneuvers, are excluded. Then, the fundamental information of orbital maneuvers for the satellites from Starlink Group 5–1 in the operational phase are statistically obtained by the analytical and numerical methods, especially the average time interval between the orbital maneuvers $\Delta\overline{S}$, the average variation of the orbital semi-major axis $\Delta\bar{a}$, the average duration $\Delta\overline{T}$, the average rate of change of the orbital semi-major axis $\dot{\bar{a}}$ and the average maneuver acceleration $|\bar{\mathbf{a}}_{th}|$ of the maneuvers, and their standard deviations ($\sigma_{\Delta\bar{S}}$, $\sigma_{\Delta\bar{a}}$, $\sigma_{\Delta\bar{T}}$, $\sigma_{\dot{\bar{a}}}$, and $\sigma_{|\bar{\mathbf{a}}_{th}|}$), see Table 3.

Similarly, excluding the "abnormal" descending maneuvers in the parking and orbit raising phases and the "abnormal" ascending maneuvers in the deorbit phase, the

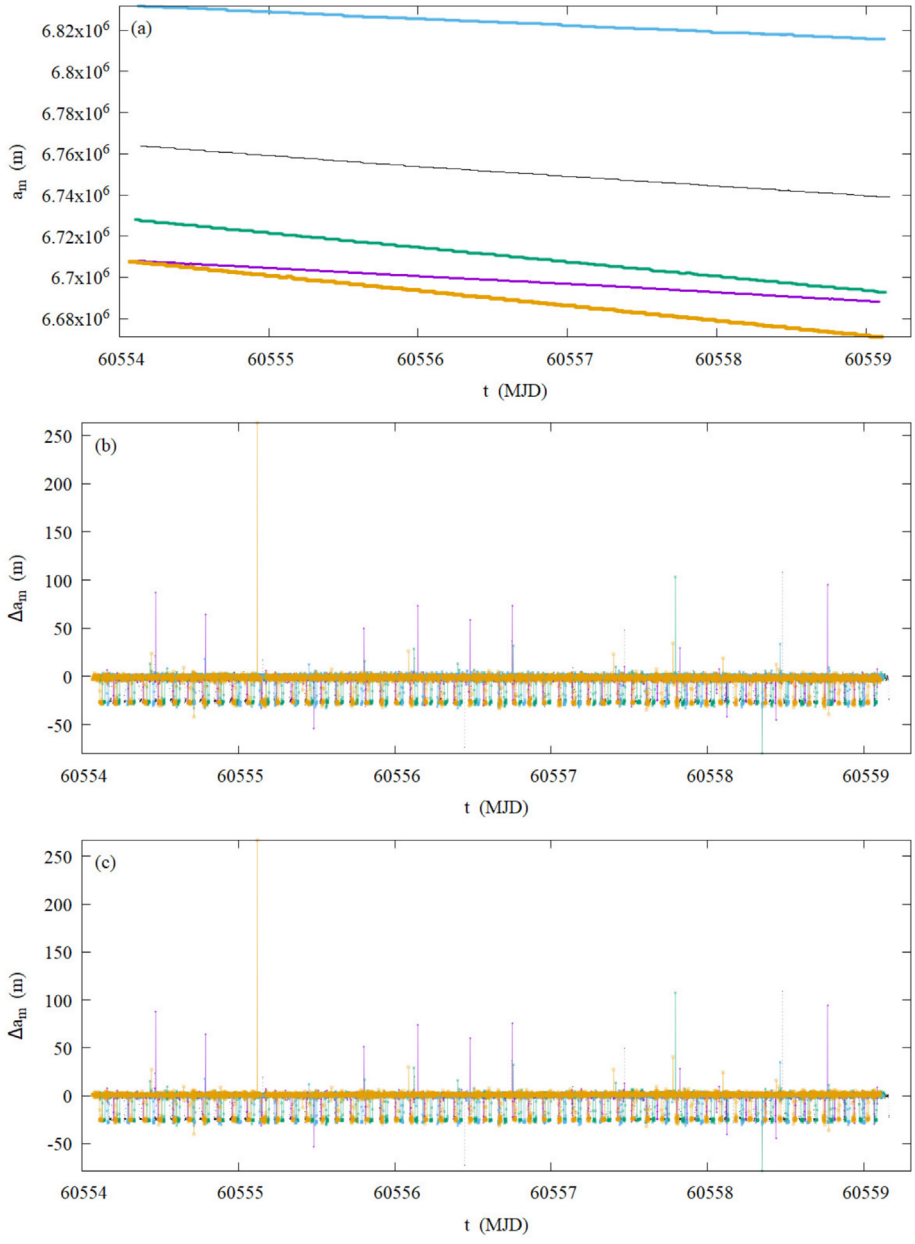


Fig. 9. For 5 satellites in the deorbit phase: (a) the evolution of the "mean orbital semi-major axis" a_m over time t retrieved by the analytical method; (b) the time series of the orbital semi-major axis variations over 1 min Δa_m caused by the maneuver force and the atmospheric drag (which can be approximated as the maneuver force), retrieved by the analytical method; (c) the time series of the orbital semi-major axis variations over 1 min Δa_m caused solely by the maneuver force, retrieved by the numerical method. The curves with different colors correspond to the different satellites.

Table 3
Retrieval results of the fundamental information of orbital maneuvers for the Starlink satellites in the operational, parking and orbit raising phases.

	Method	$\Delta\overline{S}$ (Day)	$\sigma_{\Delta\overline{S}}$ (Day)	$\Delta\bar{a}$ (m)	$\sigma_{\Delta\bar{a}}$ (m)	$\Delta\overline{T}$ (min)	$\sigma_{\Delta\overline{T}}$ (min)	$\bar{a}$ (m/s)	$\sigma_{\bar{a}}$ (m/s)	$\lvert\bar{\mathbf{a}}_{th}\rvert$ (m/s^2)	$\sigma_{\lvert\bar{\mathbf{a}}_{th}\rvert}$ (m/s^2)
Operational (Group 5–1)	Analytical	0.980	0.027	70.642	31.672	3.031	1.204	0.381	0.057	0.208E-03	0.312E-04
	Numerical	0.980	0.027	71.764	32.708	3.044	1.231	0.385	0.056	0.210E-03	0.307E-04
Parking (Group 8–3)	Analytical	0.200	0.076	242.814	134.407	12.043	6.512	0.334	0.067	0.191E-03	0.383E-04
	Numerical	0.194	0.077	255.910	147.344	11.950	6.651	0.350	0.076	0.201E-03	0.435E-04
Orbit raising (Group 10–7)	Analytical	0.032	0.002	183.678	62.446	7.761	2.567	0.394	0.043	0.226E-03	0.244E-04
	Numerical	0.032	0.002	192.833	65.466	7.852	2.571	0.408	0.041	0.234E-03	0.238E-04

corresponding fundamental information of orbital maneuvers for the satellites from Starlink Group 8–3 and Starlink Group 10–7 and for the 5 satellites Starlink-3921, Starlink-4682, Starlink-5380, Starlink-2108 and Starlink-2132 are also statistically obtained by the analytical and numerical methods, respectively (see Tables 3 and 4). Note that the obvious outliers, i.e., individual data for the satellites Starlink-32196 and Starlink 32252 (NORAD IDs 60430 and 60442), are also excluded in the statistics of the fundamental maneuver information for the satellites from Starlink Group 10–7 in the orbit raising phase; and the three-sigma rule is further activated to eliminate outliers and derive the final values in the statistics of the $\Delta\overline{S}$ and $\sigma_{\Delta\overline{S}}$ for these satellites from Starlink Group 10–7 in the orbit raising phase and for these 5 satellites in the deorbit phase.

It is seen that orbital maneuvers occur approximately every 0.980, about 0.20 and 0.032 day for the satellites from Starlink Group 5–1, Starlink Group 8–3 and Starlink Group 10–7 in the operational, parking and orbit raising phases, respectively. For these 5 satellites in the deorbit phase, their orbital altitudes are different, and the fundamental maneuver information such as the average time interval between the orbital maneuvers $\Delta\overline{S}$, the average variation of the orbital semi-major axis $\Delta\bar{a}$ and the average duration $\Delta\overline{T}$ of the maneuvers are also obviously different. The fundamental information of orbital maneuvers revealed by the analytical and numerical retrieval methods are consistent on the whole.

4. Conclusions

Taking the orbital semi-major axis as the characteristic parameter for detection of orbital maneuvers, the fundamental information of orbital maneuvers of the Starlink satellites are investigated in this paper, based on the SpaceX-released ephemerides. The analytical and numerical methods are particularly proposed to retrieve the orbital maneuver information of the Starlink satellites. Their core essences are as follows: the time series of the orbital semi-major axis variations per unit time (1 min) caused by the maneuver force and the atmospheric drag (which can be approximated as the maneuver force), are analytically retrieved through the calculations of the short-period perturbation terms in satellite orbital dynamics with the analytical formulae, and the time series of the orbital semi-major axis variations per unit time (1 min) caused solely by the maneuver force, are numerically retrieved through orbit propagation calculations, thereby revealing the orbital maneuvers of the Starlink satellites intuitively. Note that the retrieval results depend on ephemeris modeling assumptions. Furthermore, since no external maneuver truth data are publicly available, the internal consistency is directly compared between the orbital maneuver information retrieved by the analytical and numerical methods to exhibit their respective reliability.

Table 4
Retrieval results of the fundamental information of orbital maneuvers for the Starlink satellites in the deorbit phase.

NORAD ID	Method	$\Delta\overline{S}$ (Day)	$\sigma_{\Delta\overline{S}}$ (Day)	$\Delta\overline{a}$ (m)	$\sigma_{\Delta\overline{a}}$ (m)	$\Delta\overline{T}$ (min)	$\sigma_{\Delta\overline{T}}$ (min)	$\overline{\dot{a}}$ (m/s)	$\sigma_{\overline{\dot{a}}}$ (m/s)	$\lvert\overline{\mathbf{a}}_{th}\rvert$ (m/s^2)	$\sigma_{\lvert\overline{\mathbf{a}}_{th}\rvert}$ (m/s^2)
47387	Analytical	0.083	0.008	−355.476	109.251	14.661	4.613	−0.420	0.110	−0.239E-03	0.628E-04
	Numerical	0.083	0.008	−343.226	106.301	14.607	4.599	−0.407	0.112	−0.232E-03	0.639E-04
47729	Analytical	0.076	0.025	−138.446	68.049	5.762	2.728	−0.406	0.102	−0.234E-03	0.588E-04
	Numerical	0.081	0.018	−132.526	59.980	5.770	2.604	−0.396	0.094	−0.229E-03	0.541E-04
52613	Analytical	0.078	0.006	−544.345	150.192	20.589	5.730	−0.454	0.124	−0.261E-03	0.714E-04
	Numerical	0.079	0.006	−511.848	142.028	20.375	5.761	−0.432	0.123	−0.248E-03	0.708E-04
53597	Analytical	0.115	0.012	−387.977	50.385	14.447	1.884	−0.448	0.010	−0.251E-03	0.576E-05
	Numerical	0.111	0.019	−373.917	76.806	14.051	2.790	−0.437	0.044	−0.245E-03	0.246E-04
54829	Analytical	0.076	0.012	−488.212	193.507	18.525	7.264	−0.435	0.072	−0.251E-03	0.418E-04
	Numerical	0.078	0.007	−460.205	161.453	18.895	6.513	−0.410	0.052	−0.237E-03	0.301E-04

In the analytical retrieval method of this paper, the appropriate and effective amendments are made to the orbital maneuver extraction method proposed by Liu et al. (2024b): the variations of the "mean orbital semi-major axis" per unit time are used to replace the slopes of the piecewise linear fittings for the maneuver discriminations, making the maneuver discriminations more convenient and straightforward; and the short-period terms of the additional perturbation showed in the potential of the J_2 term, the third-body gravitational perturbations from the Sun and the Moon and the solar radiation pressure are further considered. Moreover, the results obtained by the analytical retrieval method demonstrate high internal consistency with the numerical retrieval method.

In the numerical retrieval method, the atmospheric drag perturbation effect is also taken into account, thereby it can better avoid the interference of the atmospheric drag perturbation on the orbital maneuver discriminations. It is worth noting that the equivalent A/m values involved in the atmospheric drag perturbation can be set empirically, with the atmospheric drag coefficient C_D being set to a basic value of 2.2, according to the fact that the orbital semi-major axis variations approach to zero in the absence of orbital maneuvers. In other words, the equivalent $C_D(A/m)$ information of the Starlink satellites involved in the atmospheric drag perturbation can be also estimated effectively by the numerical retrieval of the orbital maneuvers. In addition, because the specific dynamical model adopted in the SpaceX-released ephemerides is unknown, it can also effectively compensate model errors by taking the atmospheric drag perturbation into account and setting the involved A/m values appropriately, to make the considered dynamic model in the numerical retrieval approach the adopted dynamic model in the SpaceX-released ephemerides. Thus, in this sense, the results obtained by the numerical retrieval method are more convincing.

Using the ephemerides released by the SpaceX from September 1st to 5th, 2024 (UTC) for the selected satellites from Starlink Group 5–1 (Launch 69, 2022–177), Starlink Group 8–3 (Launch 187, 2024–141) and Starlink Group 10–7 (Launch 188, 2024–144) in the operational, parking and orbit raising phases respectively, the fundamental information of orbital maneuvers is successfully retrieved, especially the average time interval between the orbital maneuvers, the average variation of the orbital semi-major axis, the average duration, the average rate of change of the orbital semi-major axis and the average maneuver acceleration of the maneuvers, and their standard deviations. Moreover, the corresponding fundamental information of orbital maneuvers for the satellites Starlink-3921, Starlink-4682, Starlink-5380, Starlink-2108 and Starlink-2132 in the deorbit phase is also retrieved successfully. The fundamental information of orbital maneuvers revealed by the analytical and numerical retrieval methods is consistent on the whole.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The SpaceX-released ephemerides were downloaded from the website https://www.space-track.org/#publicFiles. We referred to the website https://planet4589.org/space/con/star/stats.html for information of Starlink launch statistics and orbital state history of the Starlink satellites. We thank anonymous reviewers and Associate Editor, Dr. Toshiya Hanada for making valuable comments and suggestions on our paper.

Appendix A. The detailed orbital maneuvers for the satellites Starlink-11228, Starlink-32231 and Starlink-4682

See Tables A.1, A.2, and A.3.

Table A.1
The detailed orbital maneuvers for the satellite Starlink-11228 (NORAD ID 60398) in the parking phase retrieved from the ephemerides released by the SpaceX from September 1st to 5th, 2024 (UTC).

Analytical retrieval						Numerical retrieval					
Start time	End time	ΔT (min)	Δa (m)	$\bar{\dot{a}}$ (m/s)	$\lvert\bar{\mathbf{a}}_{th}\rvert$ (m/s^2)	Start time	End time	ΔT (min)	Δa (m)	$\bar{\dot{a}}$ (m/s)	$\lvert\bar{\mathbf{a}}_{th}\rvert$ (m/s^2)
2024–09-01 04:17:42.00	2024–09-01 04:26:42.00	9	147.027	0.272	1.56E-04	2024–09-01 04:17:42.00	2024–09-01 04:26:42.00	9	160.211	0.297	1.70E-04
2024–09-01 11:05:42.00	2024–09-01 11:20:42.00	15	285.335	0.317	1.82E-04	2024–09-01 11:05:42.00	2024–09-01 11:20:42.00	15	311.261	0.346	1.98E-04
2024–09-01 16:22:42.00	2024–09-01 16:42:42.00	20	407.204	0.339	1.94E-04	2024–09-01 16:22:42.00	2024–09-01 16:42:42.00	20	435.448	0.363	2.08E-04
2024–09-01 18:59:42.00	2024–09-01 19:00:42.00	1	29.166	0.486	2.79E-04	2024–09-01 18:59:42.00	2024–09-01 19:00:42.00	1	32.163	0.536	3.07E-04
2024–09-01 22:54:42.00	2024–09-01 23:13:42.00	19	387.453	0.340	1.95E-04	2024–09-01 22:54:42.00	2024–09-01 23:13:42.00	19	403.348	0.354	2.03E-04
2024–09-02 03:55:42.00	2024–09-02 03:57:42.00	2	27.485	0.229	1.31E-04	2024–09-02 03:55:42.00	2024–09-02 03:57:42.00	2	29.971	0.250	1.43E-04
2024–09-02 10:37:42.00	2024–09-02 10:57:42.00	20	402.834	0.336	1.92E-04	2024–09-02 10:37:42.00	2024–09-02 10:57:42.00	20	431.892	0.360	2.06E-04
2024–09-02 16:01:42.00	2024–09-02 16:12:42.00	11	182.016	0.276	1.58E-04	2024–09-02 16:01:42.00	2024–09-02 16:12:42.00	11	198.594	0.301	1.72E-04
2024–09-02 22:31:42.00	2024–09-02 22:48:42.00	17	343.235	0.337	1.93E-04	2024–09-02 22:31:42.00	2024–09-02 22:48:42.00	17	362.321	0.355	2.04E-04
2024–09-03 02:57:42.00	2024–09-03 02:58:42.00	1	−12.270	−0.205	−1.17E-04	2024–09-03 02:57:42.00	2024–09-03 02:58:42.00	1	−11.525	−0.192	−1.10E-04
2024–09-03 03:26:42.00	2024–09-03 03:36:42.00	10	178.395	0.297	1.70E-04	2024–09-03 03:26:42.00	2024–09-03 03:36:42.00	10	192.615	0.321	1.84E-04
2024–09-03 10:12:42.00	2024–09-03 10:27:42.00	15	300.489	0.334	1.91E-04	2024–09-03 10:12:42.00	2024–09-03 10:27:42.00	15	329.905	0.367	2.10E-04
2024–09-03 10:28:42.00	2024–09-03 10:32:42.00	4	79.485	0.331	1.90E-04	2024–09-03 10:28:42.00	2024–09-03 10:32:42.00	4	81.112	0.338	1.94E-04
2024–09-03 16:22:42.00	2024–09-03 16:32:42.00	10	204.232	0.340	1.95E-04	2024–09-03 16:21:42.00	2024–09-03 16:32:42.00	11	227.570	0.345	1.97E-04
2024–09-03 19:20:42.00	2024–09-03 19:21:42.00	1	−15.831	−0.264	−1.51E-04	2024–09-03 19:20:42.00	2024–09-03 19:21:42.00	1	−12.817	−0.214	−1.22E-04
2024–09-03 21:19:42.00	2024–09-03 21:33:42.00	14	286.493	0.341	1.96E-04	2024–09-03 21:19:42.00	2024–09-03 21:33:42.00	14	300.871	0.358	2.06E-04
2024–09-04 02:41:42.00	2024–09-04 02:42:42.00	1	24.403	0.407	2.33E-04	2024–09-04 02:41:42.00	2024–09-04 02:42:42.00	1	26.171	0.436	2.50E-04
2024–09-04 02:58:42.00	2024–09-04 03:14:42.00	16	319.472	0.333	1.91E-04	2024–09-04 02:58:42.00	2024–09-04 03:14:42.00	16	340.611	0.355	2.03E-04
2024–09-04 09:49:42.00	2024–09-04 10:04:42.00	15	265.962	0.296	1.69E-04	2024–09-04 09:48:42.00	2024–09-04 10:04:42.00	16	301.934	0.315	1.80E-04
2024–09-04 10:41:42.00	2024–09-04 10:42:42.00	1	−104.779	−1.746	−1.00E-03	2024–09-04 10:41:42.00	2024–09-04 10:42:42.00	1	−102.632	−1.711	−9.78E-04
2024–09-04 15:08:42.00	2024–09-04 15:24:42.00	16	307.388	0.320	1.83E-04	2024–09-04 15:08:42.00	2024–09-04 15:24:42.00	16	332.349	0.346	1.98E-04
2024–09-04 18:38:42.00	2024–09-04 18:39:42.00	1	−30.153	−0.503	−2.89E-04	2024–09-04 18:38:42.00	2024–09-04 18:39:42.00	1	−28.313	−0.472	−2.71E-04
2024–09-04 22:00:42.00	2024–09-04 22:20:42.00	20	409.611	0.341	1.96E-04	2024–09-04 22:00:42.00	2024–09-04 22:20:42.00	20	435.453	0.363	2.08E-04
2024–09-05 02:49:42.00	2024–09-05 02:50:42.00	1	−14.632	−0.244	−1.40E-04	2024–09-05 02:49:42.00	2024–09-05 02:50:42.00	1	−15.116	−0.252	−1.44E-04
2024–09-05 03:20:42.00	2024–09-05 03:33:42.00	13	250.772	0.322	1.84E-04	2024–09-05 03:20:42.00	2024–09-05 03:33:42.00	13	268.430	0.344	1.97E-04
2024–09-05 09:22:42.00	2024–09-05 09:41:42.00	19	374.746	0.329	1.88E-04	2024–09-05 09:22:42.00	2024–09-05 09:41:42.00	19	403.761	0.354	2.03E-04
2024–09-05 14:45:42.00	2024–09-05 14:56:42.00	11	216.238	0.328	1.88E-04	2024–09-05 14:45:42.00	2024–09-05 14:56:42.00	11	232.584	0.352	2.02E-04
2024–09-05 19:34:42.00	2024–09-05 19:35:42.00	1	−15.251	−0.254	−1.46E-04	2024–09-05 19:34:42.00	2024–09-05 19:35:42.00	1	−15.016	−0.250	−1.43E-04
2024–09-05 20:26:42.00	2024–09-05 20:38:42.00	12	229.020	0.318	1.83E-04	2024–09-05 20:26:42.00	2024–09-05 20:38:42.00	12	246.571	0.342	1.97E-04

Table A.2
The detailed orbital maneuvers for the satellite Starlink-32231 (NORAD ID 60439) in the orbit raising phase retrieved from the ephemerides released by the SpaceX from September 1st to 5th, 2024 (UTC).

Analytical retrieval						Numerical retrieval					
Start time	End time	ΔT (min)	Δa (m)	$\bar{\dot{a}}$ (m/s)	$\lvert\bar{\mathbf{a}}_{th}\rvert$ (m/s^2)	Start time	End time	ΔT (min)	Δa (m)	$\bar{\dot{a}}$ (m/s)	$\lvert\bar{\mathbf{a}}_{th}\rvert$ (m/s^2)
2024–09-01 02:16:42.00	2024–09-01 02:29:42.00	13	293.206	0.376	2.16E-04	2024–09-01 02:16:42.00	2024–09-01 02:30:42.00	14	329.843	0.393	2.25E-04
2024–09-01 03:13:42.00	2024–09-01 03:22:42.00	9	212.276	0.393	2.26E-04	2024–09-01 03:13:42.00	2024–09-01 03:22:42.00	9	215.640	0.399	2.29E-04
2024–09-01 04:07:42.00	2024–09-01 04:17:42.00	10	223.027	0.372	2.14E-04	2024–09-01 04:07:42.00	2024–09-01 04:17:42.00	10	238.749	0.398	2.29E-04
2024–09-01 05:01:42.00	2024–09-01 05:13:42.00	12	276.866	0.385	2.21E-04	2024–09-01 05:01:42.00	2024–09-01 05:13:42.00	12	287.560	0.399	2.30E-04
2024–09-01 06:00:42.00	2024–09-01 06:03:42.00	3	62.866	0.349	2.01E-04	2024–09-01 06:00:42.00	2024–09-01 06:03:42.00	3	65.456	0.364	2.09E-04
2024–09-01 06:51:42.00	2024–09-01 07:01:42.00	10	218.374	0.364	2.09E-04	2024–09-01 06:51:42.00	2024–09-01 07:01:42.00	10	243.794	0.406	2.33E-04
2024–09-01 07:46:42.00	2024–09-01 07:55:42.00	9	218.059	0.404	2.32E-04	2024–09-01 07:46:42.00	2024–09-01 07:55:42.00	9	220.234	0.408	2.34E-04
2024–09-01 08:41:42.00	2024–09-01 08:50:42.00	9	208.455	0.386	2.22E-04	2024–09-01 08:41:42.00	2024–09-01 08:50:42.00	9	223.803	0.414	2.38E-04
2024–09-01 09:29:42.00	2024–09-01 09:30:42.00	1	10.195	0.170	9.76E-05	2024–09-01 09:29:42.00	2024–09-01 09:30:42.00	1	11.823	0.197	1.13E-04
2024–09-01 09:37:42.00	2024–09-01 09:43:42.00	6	150.175	0.417	2.40E-04	2024–09-01 09:37:42.00	2024–09-01 09:43:42.00	6	156.196	0.434	2.50E-04
2024–09-01 10:30:42.00	2024–09-01 10:39:42.00	9	223.975	0.415	2.39E-04	2024–09-01 10:30:42.00	2024–09-01 10:40:42.00	10	242.139	0.404	2.32E-04
2024–09-01 11:28:42.00	2024–09-01 11:31:42.00	3	66.549	0.370	2.12E-04	2024–09-01 11:27:42.00	2024–09-01 11:31:42.00	4	86.856	0.362	2.08E-04
2024–09-01 12:18:42.00	2024–09-01 12:30:42.00	12	303.057	0.421	2.42E-04	2024–09-01 12:18:42.00	2024–09-01 12:30:42.00	12	311.002	0.432	2.48E-04
2024–09-01 13:15:42.00	2024–09-01 13:22:42.00	7	149.237	0.355	2.04E-04	2024–09-01 13:15:42.00	2024–09-01 13:22:42.00	7	159.743	0.380	2.19E-04
2024–09-01 14:09:42.00	2024–09-01 14:18:42.00	9	215.528	0.399	2.29E-04	2024–09-01 14:09:42.00	2024–09-01 14:18:42.00	9	222.550	0.412	2.37E-04
2024–09-01 15:03:42.00	2024–09-01 15:13:42.00	10	232.718	0.388	2.23E-04	2024–09-01 15:03:42.00	2024–09-01 15:13:42.00	10	243.084	0.405	2.33E-04
2024–09-01 15:59:42.00	2024–09-01 16:06:42.00	7	140.831	0.335	1.92E-04	2024–09-01 15:59:42.00	2024–09-01 16:06:42.00	7	158.058	0.376	2.16E-04
2024–09-01 16:40:42.00	2024–09-01 16:41:42.00	1	−12.090	−0.202	−1.16E-04	2024–09-01 16:40:42.00	2024–09-01 16:41:42.00	1	−10.856	−0.181	−1.04E-04
2024–09-01 16:52:42.00	2024–09-01 17:02:42.00	10	243.164	0.405	2.33E-04	2024–09-01 16:52:42.00	2024–09-01 17:02:42.00	10	245.757	0.410	2.35E-04
2024–09-01 17:48:42.00	2024–09-01 17:55:42.00	7	167.998	0.400	2.30E-04	2024–09-01 17:48:42.00	2024–09-01 17:56:42.00	8	191.657	0.399	2.30E-04
2024–09-01 18:42:42.00	2024–09-01 18:50:42.00	8	188.808	0.393	2.26E-04	2024–09-01 18:42:42.00	2024–09-01 18:50:42.00	8	195.362	0.407	2.34E-04
2024–09-01 19:37:42.00	2024–09-01 19:45:42.00	8	195.495	0.407	2.34E-04	2024–09-01 19:37:42.00	2024–09-01 19:45:42.00	8	201.446	0.420	2.41E-04
2024–09-01 20:32:42.00	2024–09-01 20:40:42.00	8	183.894	0.383	2.20E-04	2024–09-01 20:31:42.00	2024–09-01 20:40:42.00	9	215.568	0.399	2.29E-04
2024–09-01 21:27:42.00	2024–09-01 21:33:42.00	6	138.969	0.386	2.22E-04	2024–09-01 21:27:42.00	2024–09-01 21:33:42.00	6	140.132	0.389	2.23E-04
2024–09-01 22:18:42.00	2024–09-01 22:31:42.00	13	305.347	0.391	2.25E-04	2024–09-01 22:18:42.00	2024–09-01 22:31:42.00	13	323.077	0.414	2.38E-04
2024–09-01 23:17:42.00	2024–09-01 23:22:42.00	5	110.308	0.368	2.11E-04	2024–09-01 23:17:42.00	2024–09-01 23:22:42.00	5	114.826	0.383	2.20E-04
2024–09-02 00:10:42.00	2024–09-02 00:19:42.00	9	217.811	0.403	2.32E-04	2024–09-02 00:10:42.00	2024–09-02 00:19:42.00	9	226.295	0.419	2.41E-04
2024–09-02 01:05:42.00	2024–09-02 01:13:42.00	8	189.541	0.395	2.27E-04	2024–09-02 01:04:42.00	2024–09-02 01:13:42.00	9	216.571	0.401	2.30E-04
2024–09-02 01:59:42.00	2024–09-02 02:08:42.00	9	212.079	0.393	2.25E-04	2024–09-02 01:59:42.00	2024–09-02 02:08:42.00	9	214.995	0.398	2.28E-04
2024–09-02 02:55:42.00	2024–09-02 03:02:42.00	7	156.101	0.372	2.14E-04	2024–09-02 02:55:42.00	2024–09-02 03:02:42.00	7	167.195	0.398	2.29E-04
2024–09-02 03:47:42.00	2024–09-02 03:59:42.00	12	282.706	0.393	2.25E-04	2024–09-02 03:47:42.00	2024–09-02 03:59:42.00	12	293.114	0.407	2.34E-04
2024–09-02 04:47:42.00	2024–09-02 04:48:42.00	1	24.500	0.408	2.35E-04	2024–09-02 04:47:42.00	2024–09-02 04:48:42.00	1	25.477	0.425	2.44E-04
2024–09-02 05:36:42.00	2024–09-02 05:49:42.00	13	293.787	0.377	2.16E-04	2024–09-02 05:36:42.00	2024–09-02 05:49:42.00	13	319.325	0.409	2.35E-04
2024–09-02 06:34:42.00	2024–09-02 06:39:42.00	5	121.530	0.405	2.32E-04	2024–09-02 06:34:42.00	2024–09-02 06:39:42.00	5	122.102	0.407	2.33E-04
2024–09-02 07:26:42.00	2024–09-02 07:37:42.00	11	262.649	0.398	2.29E-04	2024–09-02 07:26:42.00	2024–09-02 07:37:42.00	11	278.233	0.422	2.42E-04
2024–09-02 08:23:42.00	2024–09-02 08:30:42.00	7	155.723	0.371	2.13E-04	2024–09-02 08:23:42.00	2024–09-02 08:30:42.00	7	160.890	0.383	2.20E-04
2024–09-02 09:16:42.00	2024–09-02 09:26:42.00	10	239.942	0.400	2.30E-04	2024–09-02 09:16:42.00	2024–09-02 09:26:42.00	10	247.025	0.412	2.37E-04
2024–09-02 09:51:42.00	2024–09-02 09:52:42.00	1	−12.204	−0.203	−1.17E-04	-	-	-	-	-	-
2024–09-02 10:13:42.00	2024–09-02 10:18:42.00	5	111.217	0.371	2.13E-04	2024–09-02 10:13:42.00	2024–09-02 10:18:42.00	5	123.366	0.411	2.36E-04
2024–09-02 11:05:42.00	2024–09-02 11:16:42.00	11	264.724	0.401	2.30E-04	2024–09-02 11:05:42.00	2024–09-02 11:16:42.00	11	269.283	0.408	2.34E-04
2024–09-02 12:02:42.00	2024–09-02 12:08:42.00	6	146.575	0.407	2.34E-04	2024–09-02 12:02:42.00	2024–09-02 12:08:42.00	6	155.573	0.432	2.48E-04
2024–09-02 12:55:42.00	2024–09-02 13:04:42.00	9	225.175	0.417	2.39E-04	2024–09-02 12:55:42.00	2024–09-02 13:04:42.00	9	232.704	0.431	2.48E-04
2024–09-02 13:50:42.00	2024–09-02 13:58:42.00	8	200.076	0.417	2.40E-04	2024–09-02 13:50:42.00	2024–09-02 13:58:42.00	8	206.032	0.429	2.47E-04
2024–09-02 14:46:42.00	2024–09-02 14:52:42.00	6	132.626	0.368	2.11E-04	2024–09-02 14:46:42.00	2024–09-02 14:52:42.00	6	145.790	0.405	2.32E-04
2024–09-02 15:39:42.00	2024–09-02 15:48:42.00	9	210.830	0.390	2.24E-04	2024–09-02 15:39:42.00	2024–09-02 15:48:42.00	9	211.899	0.392	2.25E-04

Table A.2 (*continued*)

Analytical retrieval						Numerical retrieval					
Start time	End time	ΔT (min)	Δa (m)	$\bar{\dot{a}}$ (m/s)	$\lvert\bar{\mathbf{a}}_{th}\rvert$ (m/s^2)	Start time	End time	ΔT (min)	Δa (m)	$\bar{\dot{a}}$ (m/s)	$\lvert\bar{\mathbf{a}}_{th}\rvert$ (m/s^2)
2024–09-02 15:56:42.00	2024–09-02 15:57:42.00	1	23.337	0.389	2.23E-04	2024–09-02 15:56:42.00	2024–09-02 15:57:42.00	1	24.730	0.412	2.37E-04
2024–09-02 16:35:42.00	2024–09-02 16:42:42.00	7	155.237	0.370	2.12E-04	2024–09-02 16:35:42.00	2024–09-02 16:42:42.00	7	166.263	0.396	2.28E-04
2024–09-02 17:28:42.00	2024–09-02 17:38:42.00	10	223.338	0.372	2.14E-04	2024–09-02 17:28:42.00	2024–09-02 17:38:42.00	10	231.942	0.387	2.22E-04
2024–09-02 18:24:42.00	2024–09-02 18:31:42.00	7	165.622	0.394	2.27E-04	2024–09-02 18:24:42.00	2024–09-02 18:31:42.00	7	170.094	0.405	2.33E-04
2024–09-02 19:19:42.00	2024–09-02 19:26:42.00	7	161.587	0.385	2.21E-04	2024–09-02 19:18:42.00	2024–09-02 19:26:42.00	8	187.793	0.391	2.24E-04
2024–09-02 20:13:42.00	2024–09-02 20:21:42.00	8	183.680	0.383	2.19E-04	2024–09-02 20:13:42.00	2024–09-02 20:21:42.00	8	185.215	0.386	2.21E-04
2024–09-02 21:06:42.00	2024–09-02 21:17:42.00	11	265.842	0.403	2.31E-04	2024–09-02 21:06:42.00	2024–09-02 21:17:42.00	11	280.760	0.425	2.44E-04
2024–09-02 22:04:42.00	2024–09-02 22:09:42.00	5	116.976	0.390	2.24E-04	2024–09-02 22:04:42.00	2024–09-02 22:09:42.00	5	120.636	0.402	2.31E-04
2024–09-02 22:56:42.00	2024–09-02 23:06:42.00	10	224.350	0.374	2.15E-04	2024–09-02 22:56:42.00	2024–09-02 23:06:42.00	10	232.074	0.387	2.22E-04
2024–09-02 23:52:42.00	2024–09-03 00:00:42.00	8	192.950	0.402	2.31E-04	2024–09-02 23:52:42.00	2024–09-03 00:00:42.00	8	208.977	0.435	2.49E-04
2024–09-03 00:48:42.00	2024–09-03 00:54:42.00	6	132.789	0.369	2.11E-04	2024–09-03 00:48:42.00	2024–09-03 00:54:42.00	6	132.584	0.368	2.11E-04
2024–09-03 01:41:42.00	2024–09-03 01:50:42.00	9	211.883	0.392	2.25E-04	2024–09-03 01:41:42.00	2024–09-03 01:50:42.00	9	225.237	0.417	2.40E-04
2024–09-03 02:34:42.00	2024–09-03 02:46:42.00	12	302.487	0.420	2.41E-04	2024–09-03 02:34:42.00	2024–09-03 02:46:42.00	12	313.288	0.435	2.50E-04
2024–09-03 03:34:42.00	2024–09-03 03:35:42.00	1	25.581	0.426	2.45E-04	2024–09-03 03:34:42.00	2024–09-03 03:35:42.00	1	26.067	0.434	2.50E-04
2024–09-03 04:23:42.00	2024–09-03 04:35:42.00	12	277.741	0.386	2.21E-04	2024–09-03 04:23:42.00	2024–09-03 04:35:42.00	12	302.079	0.420	2.40E-04
2024–09-03 05:22:42.00	2024–09-03 05:26:42.00	4	92.900	0.387	2.22E-04	2024–09-03 05:22:42.00	2024–09-03 05:26:42.00	4	92.520	0.385	2.21E-04
2024–09-03 06:12:42.00	2024–09-03 06:25:42.00	13	307.704	0.394	2.27E-04	2024–09-03 06:12:42.00	2024–09-03 06:25:42.00	13	322.886	0.414	2.38E-04
2024–09-03 07:10:42.00	2024–09-03 07:17:42.00	7	166.541	0.397	2.27E-04	2024–09-03 07:10:42.00	2024–09-03 07:17:42.00	7	171.005	0.407	2.34E-04
2024–09-03 08:05:42.00	2024–09-03 08:12:42.00	7	174.913	0.416	2.39E-04	2024–09-03 08:05:42.00	2024–09-03 08:12:42.00	7	178.866	0.426	2.45E-04
2024–09-03 09:00:42.00	2024–09-03 09:06:42.00	6	142.513	0.396	2.27E-04	2024–09-03 09:00:42.00	2024–09-03 09:06:42.00	6	156.748	0.435	2.49E-04
2024–09-03 09:53:42.00	2024–09-03 09:58:42.00	5	123.993	0.413	2.37E-04	2024–09-03 09:53:42.00	2024–09-03 09:58:42.00	5	127.037	0.423	2.42E-04
2024–09-03 09:59:42.00	2024–09-03 10:03:42.00	4	102.419	0.427	2.45E-04	2024–09-03 09:59:42.00	2024–09-03 10:03:42.00	4	101.971	0.425	2.43E-04
2024–09-03 10:49:42.00	2024–09-03 10:56:42.00	7	153.015	0.364	2.09E-04	2024–09-03 10:49:42.00	2024–09-03 10:56:42.00	7	163.728	0.390	2.24E-04
2024–09-03 11:41:42.00	2024–09-03 11:54:42.00	13	307.372	0.394	2.26E-04	2024–09-03 11:41:42.00	2024–09-03 11:54:42.00	13	316.140	0.405	2.33E-04
2024–09-03 12:39:42.00	2024–09-03 12:45:42.00	6	135.314	0.376	2.16E-04	2024–09-03 12:39:42.00	2024–09-03 12:45:42.00	6	140.552	0.390	2.24E-04
2024–09-03 13:33:42.00	2024–09-03 13:41:42.00	8	174.809	0.364	2.09E-04	2024–09-03 13:33:42.00	2024–09-03 13:41:42.00	8	191.021	0.398	2.28E-04
2024–09-03 14:28:42.00	2024–09-03 14:34:42.00	6	146.781	0.408	2.34E-04	2024–09-03 14:28:42.00	2024–09-03 14:34:42.00	6	144.787	0.402	2.30E-04
2024–09-03 15:21:42.00	2024–09-03 15:31:42.00	10	225.742	0.376	2.16E-04	2024–09-03 15:21:42.00	2024–09-03 15:31:42.00	10	239.184	0.399	2.29E-04
2024–09-03 16:17:42.00	2024–09-03 16:25:42.00	8	191.694	0.399	2.29E-04	2024–09-03 16:17:42.00	2024–09-03 16:25:42.00	8	198.243	0.413	2.37E-04
2024–09-03 17:11:42.00	2024–09-03 17:20:42.00	9	206.396	0.382	2.19E-04	2024–09-03 17:11:42.00	2024–09-03 17:20:42.00	9	213.558	0.395	2.27E-04
2024–09-03 18:07:42.00	2024–09-03 18:13:42.00	6	133.434	0.371	2.12E-04	2024–09-03 18:07:42.00	2024–09-03 18:14:42.00	7	157.488	0.375	2.15E-04
2024–09-03 19:01:42.00	2024–09-03 19:10:42.00	9	223.503	0.414	2.37E-04	2024–09-03 19:00:42.00	2024–09-03 19:10:42.00	10	234.017	0.390	2.23E-04
2024–09-03 19:19:42.00	2024–09-03 19:20:42.00	1	−24.193	−0.403	−2.31E-04	2024–09-03 19:19:42.00	2024–09-03 19:20:42.00	1	−23.028	−0.384	−2.20E-04
2024–09-03 19:54:42.00	2024–09-03 20:05:42.00	11	256.473	0.389	2.23E-04	2024–09-03 19:54:42.00	2024–09-03 20:05:42.00	11	269.230	0.408	2.34E-04
2024–09-03 20:51:42.00	2024–09-03 20:58:42.00	7	171.685	0.409	2.34E-04	2024–09-03 20:51:42.00	2024–09-03 20:58:42.00	7	176.402	0.420	2.41E-04
2024–09-03 21:46:42.00	2024–09-03 21:52:42.00	6	148.598	0.413	2.37E-04	2024–09-03 21:46:42.00	2024–09-03 21:53:42.00	7	162.826	0.388	2.23E-04
2024–09-03 22:39:42.00	2024–09-03 22:49:42.00	10	222.426	0.371	2.12E-04	2024–09-03 22:39:42.00	2024–09-03 22:49:42.00	10	241.684	0.403	2.31E-04
2024–09-03 23:35:42.00	2024–09-03 23:42:42.00	7	176.728	0.421	2.41E-04	2024–09-03 23:35:42.00	2024–09-03 23:42:42.00	7	174.550	0.416	2.38E-04
2024–09-04 00:29:42.00	2024–09-04 00:37:42.00	8	181.906	0.379	2.18E-04	2024–09-04 00:29:42.00	2024–09-04 00:37:42.00	8	192.469	0.401	2.30E-04
2024–09-04 01:23:42.00	2024–09-04 01:33:42.00	10	238.352	0.397	2.28E-04	2024–09-04 01:23:42.00	2024–09-04 01:33:42.00	10	248.716	0.415	2.38E-04
2024–09-04 02:20:42.00	2024–09-04 02:26:42.00	6	132.206	0.367	2.11E-04	2024–09-04 02:20:42.00	2024–09-04 02:26:42.00	6	136.017	0.378	2.17E-04
2024–09-04 03:14:42.00	2024–09-04 03:22:42.00	8	187.145	0.390	2.23E-04	2024–09-04 03:14:42.00	2024–09-04 03:22:42.00	8	204.113	0.425	2.43E-04
2024–09-04 04:09:42.00	2024–09-04 04:16:42.00	7	184.119	0.438	2.51E-04	2024–09-04 04:08:42.00	2024–09-04 04:16:42.00	8	193.487	0.403	2.31E-04
2024–09-04 05:02:42.00	2024–09-04 05:12:42.00	10	230.888	0.385	2.21E-04	2024–09-04 05:02:42.00	2024–09-04 05:12:42.00	10	243.482	0.406	2.33E-04
2024–09-04 05:59:42.00	2024–09-04 06:05:42.00	6	149.585	0.416	2.38E-04	2024–09-04 05:59:42.00	2024–09-04 06:05:42.00	6	151.469	0.421	2.42E-04
2024–09-04 06:52:42.00	2024–09-04 07:02:42.00	10	242.793	0.405	2.32E-04	2024–09-04 06:52:42.00	2024–09-04 07:02:42.00	10	251.115	0.419	2.40E-04
2024–09-04 07:48:42.00	2024–09-04 07:55:42.00	7	159.432	0.380	2.17E-04	2024–09-04 07:48:42.00	2024–09-04 07:55:42.00	7	175.530	0.418	2.39E-04

(*continued on next page*)

Table A.2 (*continued*)

Analytical retrieval						Numerical retrieval					
Start time	End time	ΔT (min)	Δa (m)	$\bar{a}$ (m/s)	$\lvert\bar{\mathbf{a}}_{th}\rvert$ (m/s^2)	Start time	End time	ΔT (min)	Δa (m)	$\bar{a}$ (m/s)	$\lvert\bar{\mathbf{a}}_{th}\rvert$ (m/s^2)
2024–09-04 08:43:42.00	2024–09-04 08:50:42.00	7	177.522	0.423	2.42E-04	2024–09-04 08:43:42.00	2024–09-04 08:50:42.00	7	176.711	0.421	2.41E-04
2024–09-04 09:36:42.00	2024–09-04 09:46:42.00	10	219.496	0.366	2.10E-04	2024–09-04 09:36:42.00	2024–09-04 09:46:42.00	10	233.181	0.389	2.23E-04
2024–09-04 09:51:42.00	2024–09-04 09:52:42.00	1	14.239	0.237	1.36E-04	2024–09-04 09:51:42.00	2024–09-04 09:52:42.00	1	15.174	0.253	1.45E-04
2024–09-04 10:31:42.00	2024–09-04 10:40:42.00	9	222.220	0.412	2.36E-04	2024–09-04 10:31:42.00	2024–09-04 10:40:42.00	9	229.937	0.426	2.44E-04
2024–09-04 11:29:42.00	2024–09-04 11:32:42.00	3	76.785	0.427	2.45E-04	2024–09-04 11:29:42.00	2024–09-04 11:33:42.00	4	88.850	0.370	2.13E-04
2024–09-04 12:22:42.00	2024–09-04 12:29:42.00	7	162.478	0.387	2.22E-04	2024–09-04 12:22:42.00	2024–09-04 12:29:42.00	7	178.576	0.425	2.43E-04
2024–09-04 13:14:42.00	2024–09-04 13:26:42.00	12	289.086	0.402	2.30E-04	2024–09-04 13:14:42.00	2024–09-04 13:26:42.00	12	292.666	0.406	2.33E-04
2024–09-04 14:11:42.00	2024–09-04 14:19:42.00	8	182.148	0.379	2.18E-04	2024–09-04 14:11:42.00	2024–09-04 14:19:42.00	8	194.394	0.405	2.32E-04
2024–09-04 15:06:42.00	2024–09-04 15:13:42.00	7	177.502	0.423	2.42E-04	2024–09-04 15:06:42.00	2024–09-04 15:13:42.00	7	184.579	0.439	2.52E-04
2024–09-04 16:00:42.00	2024–09-04 16:08:42.00	8	193.250	0.403	2.31E-04	2024–09-04 16:00:42.00	2024–09-04 16:08:42.00	8	199.275	0.415	2.38E-04
2024–09-04 16:56:42.00	2024–09-04 17:02:42.00	6	77.480	0.215	1.23E-04	2024–09-04 16:56:42.00	2024–09-04 17:02:42.00	6	91.372	0.254	1.45E-04
2024–09-04 17:47:42.00	2024–09-04 18:00:42.00	13	317.924	0.408	2.33E-04	2024–09-04 17:47:42.00	2024–09-04 18:01:42.00	14	335.340	0.399	2.28E-04
2024–09-04 18:45:42.00	2024–09-04 18:52:42.00	7	166.030	0.395	2.27E-04	2024–09-04 18:45:42.00	2024–09-04 18:52:42.00	7	175.177	0.417	2.39E-04
2024–09-04 19:39:42.00	2024–09-04 19:48:42.00	9	226.277	0.419	2.40E-04	2024–09-04 19:39:42.00	2024–09-04 19:48:42.00	9	232.248	0.430	2.47E-04
2024–09-04 20:35:42.00	2024–09-04 20:41:42.00	6	147.842	0.411	2.36E-04	2024–09-04 20:35:42.00	2024–09-04 20:42:42.00	7	163.019	0.388	2.23E-04
2024–09-04 21:29:42.00	2024–09-04 21:37:42.00	8	177.500	0.370	2.12E-04	2024–09-04 21:29:42.00	2024–09-04 21:37:42.00	8	195.999	0.408	2.34E-04
2024–09-04 22:25:42.00	2024–09-04 22:31:42.00	6	154.010	0.428	2.45E-04	2024–09-04 22:24:42.00	2024–09-04 22:31:42.00	7	161.913	0.386	2.20E-04
2024–09-04 23:17:42.00	2024–09-04 23:28:42.00	11	246.249	0.373	2.14E-04	2024–09-04 23:17:42.00	2024–09-04 23:28:42.00	11	260.904	0.395	2.27E-04
2024–09-05 00:14:42.00	2024–09-05 00:21:42.00	7	169.153	0.403	2.31E-04	2024–09-05 00:14:42.00	2024–09-05 00:21:42.00	7	175.489	0.418	2.40E-04
2024–09-05 01:08:42.00	2024–09-05 01:17:42.00	9	210.501	0.390	2.24E-04	2024–09-05 01:08:42.00	2024–09-05 01:17:42.00	9	217.736	0.403	2.31E-04
2024–09-05 02:05:42.00	2024–09-05 02:10:42.00	5	105.291	0.351	2.01E-04	2024–09-05 02:04:42.00	2024–09-05 02:10:42.00	6	127.266	0.354	2.02E-04
2024–09-05 02:56:42.00	2024–09-05 03:07:42.00	11	270.967	0.411	2.35E-04	2024–09-05 02:56:42.00	2024–09-05 03:07:42.00	11	272.611	0.413	2.36E-04
2024–09-05 03:52:42.00	2024–09-05 04:01:42.00	9	221.668	0.410	2.35E-04	2024–09-05 03:52:42.00	2024–09-05 04:01:42.00	9	233.128	0.432	2.48E-04
2024–09-05 04:49:42.00	2024–09-05 04:54:42.00	5	118.545	0.395	2.26E-04	2024–09-05 04:49:42.00	2024–09-05 04:54:42.00	5	120.026	0.400	2.29E-04
2024–09-05 05:42:42.00	2024–09-05 05:50:42.00	8	197.896	0.412	2.36E-04	2024–09-05 05:42:42.00	2024–09-05 05:50:42.00	8	203.807	0.425	2.44E-04
2024–09-05 06:37:42.00	2024–09-05 06:45:42.00	8	186.989	0.390	2.23E-04	2024–09-05 06:37:42.00	2024–09-05 06:45:42.00	8	204.417	0.426	2.43E-04
2024–09-05 07:32:42.00	2024–09-05 07:40:42.00	8	185.710	0.387	2.21E-04	2024–09-05 07:32:42.00	2024–09-05 07:39:42.00	7	174.873	0.416	2.38E-04
2024–09-05 08:26:42.00	2024–09-05 08:35:42.00	9	218.843	0.405	2.32E-04	2024–09-05 08:26:42.00	2024–09-05 08:35:42.00	9	229.785	0.426	2.44E-04
2024–09-05 09:21:42.00	2024–09-05 09:29:42.00	8	196.416	0.409	2.34E-04	2024–09-05 09:21:42.00	2024–09-05 09:29:42.00	8	203.436	0.424	2.43E-04
2024–09-05 10:16:42.00	2024–09-05 10:24:42.00	8	188.100	0.392	2.25E-04	2024–09-05 10:16:42.00	2024–09-05 10:24:42.00	8	193.664	0.403	2.31E-04
2024–09-05 11:13:42.00	2024–09-05 11:18:42.00	5	108.275	0.361	2.07E-04	2024–09-05 11:13:42.00	2024–09-05 11:18:42.00	5	118.366	0.395	2.26E-04
2024–09-05 12:05:42.00	2024–09-05 12:15:42.00	10	246.818	0.411	2.35E-04	2024–09-05 12:05:42.00	2024–09-05 12:15:42.00	10	249.203	0.415	2.37E-04
2024–09-05 13:00:42.00	2024–09-05 13:10:42.00	10	232.872	0.388	2.22E-04	2024–09-05 13:00:42.00	2024–09-05 13:10:42.00	10	245.641	0.409	2.35E-04
2024–09-05 13:57:42.00	2024–09-05 14:02:42.00	5	115.062	0.384	2.20E-04	2024–09-05 13:57:42.00	2024–09-05 14:02:42.00	5	119.043	0.397	2.28E-04
2024–09-05 14:49:42.00	2024–09-05 15:00:42.00	11	269.643	0.409	2.34E-04	2024–09-05 14:49:42.00	2024–09-05 15:00:42.00	11	279.726	0.424	2.43E-04
2024–09-05 15:47:42.00	2024–09-05 15:52:42.00	5	111.997	0.373	2.14E-04	2024–09-05 15:47:42.00	2024–09-05 15:52:42.00	5	122.697	0.409	2.34E-04
2024–09-05 16:40:42.00	2024–09-05 16:48:42.00	8	199.596	0.416	2.38E-04	2024–09-05 16:40:42.00	2024–09-05 16:48:42.00	8	199.268	0.415	2.37E-04
2024–09-05 17:35:42.00	2024–09-05 17:43:42.00	8	189.451	0.395	2.26E-04	2024–09-05 17:35:42.00	2024–09-05 17:43:42.00	8	199.447	0.416	2.38E-04
2024–09-05 17:59:42.00	2024–09-05 18:00:42.00	1	−14.758	−0.246	−1.41E-04	2024–09-05 17:59:42.00	2024–09-05 18:00:42.00	1	−13.819	−0.230	−1.32E-04
2024–09-05 18:29:42.00	2024–09-05 18:38:42.00	9	223.867	0.415	2.37E-04	2024–09-05 18:29:42.00	2024–09-05 18:38:42.00	9	229.000	0.424	2.43E-04
2024–09-05 19:24:42.00	2024–09-05 19:32:42.00	8	187.267	0.390	2.24E-04	2024–09-05 19:24:42.00	2024–09-05 19:32:42.00	8	193.301	0.403	2.31E-04
2024–09-05 20:19:42.00	2024–09-05 20:28:42.00	9	199.983	0.370	2.12E-04	2024–09-05 20:19:42.00	2024–09-05 20:28:42.00	9	219.022	0.406	2.32E-04
2024–09-05 21:15:42.00	2024–09-05 21:21:42.00	6	153.874	0.427	2.44E-04	2024–09-05 21:15:42.00	2024–09-05 21:21:42.00	6	152.903	0.425	2.43E-04
2024–09-05 22:07:42.00	2024–09-05 22:18:42.00	11	257.829	0.391	2.24E-04	2024–09-05 22:07:42.00	2024–09-05 22:18:42.00	11	270.660	0.410	2.35E-04
2024–09-05 23:04:42.00	2024–09-05 23:11:42.00	7	168.277	0.401	2.29E-04	2024–09-05 23:04:42.00	2024–09-05 23:11:42.00	7	174.739	0.416	2.38E-04
2024–09-05 23:59:42.00	2024–09-06 00:06:42.00	7	157.199	0.374	2.15E-04	2024–09-05 23:59:42.00	2024–09-06 00:06:42.00	7	160.891	0.383	2.20E-04
2024–09-06 00:53:42.00	2024–09-06 01:02:42.00	9	220.331	0.408	2.33E-04	2024–09-06 00:53:42.00	2024–09-06 01:02:42.00	9	237.319	0.439	2.51E-04

Table A.3
The detailed orbital maneuvers for the satellite Starlink-4682 (NORAD ID 53597) in the deorbit phase retrieved from the ephemerides released by the SpaceX from September 1st to 5th, 2024 (UTC).

Analytical retrieval						Numerical retrieval					
Start time	End time	ΔT (min)	Δa (m)	$\bar{\dot{a}}$ (m/s)	$\lvert\bar{\mathbf{a}}_{th}\rvert$ (m/s^2)	Start time	End time	ΔT (min)	Δa (m)	$\bar{\dot{a}}$ (m/s)	$\lvert\bar{\mathbf{a}}_{th}\rvert$ (m/s^2)
2024–09-01 03:52:42.00	2024–09-01 04:04:42.00	12	−323.106	−0.449	−2.51E-04	2024–09-01 03:52:42.00	2024–09-01 04:04:42.00	12	−320.889	−0.446	−2.50E-04
2024–09-01 06:58:42.00	2024–09-01 07:13:42.00	15	−412.745	−0.459	−2.57E-04	2024–09-01 06:58:42.00	2024–09-01 07:13:42.00	15	−410.114	−0.456	−2.55E-04
2024–09-01 10:52:42.00	2024–09-01 11:06:42.00	14	−378.102	−0.450	−2.52E-04	2024–09-01 10:52:42.00	2024–09-01 11:06:42.00	14	−374.383	−0.446	−2.49E-04
2024–09-01 13:59:42.00	2024–09-01 14:13:42.00	14	−384.642	−0.458	−2.56E-04	2024–09-01 13:59:42.00	2024–09-01 14:13:42.00	14	−381.078	−0.454	−2.54E-04
2024–09-01 17:05:42.00	2024–09-01 17:21:42.00	16	−421.289	−0.439	−2.45E-04	2024–09-01 17:05:42.00	2024–09-01 17:21:42.00	16	−417.556	−0.435	−2.43E-04
2024–09-01 18:49:42.00	2024–09-01 18:50:42.00	1	18.469	0.308	1.72E-04	2024–09-01 18:49:42.00	2024–09-01 18:50:42.00	1	17.453	0.291	1.63E-04
2024–09-01 20:59:42.00	2024–09-01 21:14:42.00	15	−407.565	−0.453	−2.54E-04	2024–09-01 20:59:42.00	2024–09-01 21:14:42.00	15	−402.275	−0.447	−2.51E-04
2024–09-02 00:06:42.00	2024–09-02 00:22:42.00	16	−417.491	−0.435	−2.44E-04	2024–09-02 00:06:42.00	2024–09-02 00:22:42.00	16	−413.384	−0.431	−2.41E-04
2024–09-02 03:14:42.00	2024–09-02 03:30:42.00	16	−418.455	−0.436	−2.44E-04	2024–09-02 03:14:42.00	2024–09-02 03:30:42.00	16	−416.269	−0.434	−2.43E-04
2024–09-02 05:34:42.00	2024–09-02 05:49:42.00	15	−404.669	−0.450	−2.51E-04	2024–09-02 05:34:42.00	2024–09-02 05:49:42.00	15	−404.031	−0.449	−2.51E-04
2024–09-02 08:41:42.00	2024–09-02 08:56:42.00	15	−410.250	−0.456	−2.55E-04	2024–09-02 08:41:42.00	2024–09-02 08:56:42.00	15	−406.049	−0.451	−2.53E-04
2024–09-02 10:50:42.00	2024–09-02 10:51:42.00	1	12.808	0.213	1.19E-04	2024–09-02 10:50:42.00	2024–09-02 10:51:42.00	1	11.861	0.198	1.10E-04
2024–09-02 11:48:42.00	2024–09-02 12:03:42.00	15	−405.589	−0.451	−2.52E-04	2024–09-02 11:48:42.00	2024–09-02 12:03:42.00	15	−401.618	−0.446	−2.50E-04
2024–09-02 14:55:42.00	2024–09-02 15:09:42.00	14	−369.518	−0.440	−2.46E-04	2024–09-02 14:55:42.00	2024–09-02 15:09:42.00	14	−365.887	−0.436	−2.44E-04
2024–09-02 18:48:42.00	2024–09-02 19:04:42.00	16	−415.316	−0.433	−2.42E-04	2024–09-02 18:48:42.00	2024–09-02 19:04:42.00	16	−410.322	−0.427	−2.40E-04
2024–09-02 21:55:42.00	2024–09-02 22:11:42.00	16	−416.152	−0.433	−2.43E-04	2024–09-02 21:55:42.00	2024–09-02 22:11:42.00	16	−411.470	−0.429	−2.41E-04
2024–09-03 01:02:42.00	2024–09-03 01:18:42.00	16	−414.542	−0.432	−2.42E-04	2024–09-03 01:03:42.00	2024–09-03 01:18:42.00	15	−400.764	−0.445	−2.50E-04
-	-	-	-	-	-	2024–09-03 03:13:42.00	2024–09-03 03:14:42.00	1	−10.726	−0.179	−9.99E-05
2024–09-03 04:13:42.00	2024–09-03 04:23:42.00	10	−265.634	−0.443	−2.48E-04	2024–09-03 04:13:42.00	2024–09-03 04:23:42.00	10	−264.733	−0.441	−2.48E-04
2024–09-03 06:32:42.00	2024–09-03 06:42:42.00	10	−270.733	−0.451	−2.53E-04	2024–09-03 06:32:42.00	2024–09-03 06:43:42.00	11	−278.235	−0.422	−2.36E-04
2024–09-03 09:36:42.00	2024–09-03 09:52:42.00	16	−424.093	−0.442	−2.47E-04	2024–09-03 09:37:42.00	2024–09-03 09:52:42.00	15	−409.241	−0.455	−2.55E-04
2024–09-03 12:44:42.00	2024–09-03 12:58:42.00	14	−384.706	−0.458	−2.56E-04	2024–09-03 12:44:42.00	2024–09-03 12:58:42.00	14	−380.502	−0.453	−2.54E-04
2024–09-03 15:50:42.00	2024–09-03 16:06:42.00	16	−419.158	−0.437	−2.44E-04	2024–09-03 15:50:42.00	2024–09-03 16:06:42.00	16	−415.775	−0.433	−2.43E-04
2024–09-03 19:44:42.00	2024–09-03 19:59:42.00	15	−413.551	−0.460	−2.58E-04	2024–09-03 19:44:42.00	2024–09-03 19:59:42.00	15	−409.225	−0.455	−2.55E-04
2024–09-03 22:51:42.00	2024–09-03 23:06:42.00	15	−408.090	−0.453	−2.54E-04	2024–09-03 22:51:42.00	2024–09-03 23:06:42.00	15	−404.412	−0.449	−2.52E-04
2024–09-04 01:58:42.00	2024–09-04 02:13:42.00	15	−407.701	−0.453	−2.54E-04	2024–09-04 01:58:42.00	2024–09-04 02:13:42.00	15	−405.100	−0.450	−2.53E-04
2024–09-04 04:18:42.00	2024–09-04 04:33:42.00	15	−414.023	−0.460	−2.58E-04	2024–09-04 04:18:42.00	2024–09-04 04:33:42.00	15	−410.128	−0.456	−2.56E-04
2024–09-04 07:28:42.00	2024–09-04 07:36:42.00	8	−216.847	−0.452	−2.53E-04	2024–09-04 07:28:42.00	2024–09-04 07:36:42.00	8	−214.554	−0.447	−2.51E-04
2024–09-04 10:31:42.00	2024–09-04 10:46:42.00	15	−402.998	−0.448	−2.51E-04	2024–09-04 10:31:42.00	2024–09-04 10:46:42.00	15	−397.249	−0.441	−2.48E-04
2024–09-04 14:24:42.00	2024–09-04 14:40:42.00	16	−417.786	−0.435	−2.44E-04	2024–09-04 14:24:42.00	2024–09-04 14:40:42.00	16	−412.491	−0.430	−2.41E-04
2024–09-04 17:31:42.00	2024–09-04 17:46:42.00	15	−409.821	−0.455	−2.56E-04	2024–09-04 17:31:42.00	2024–09-04 17:46:42.00	15	−404.148	−0.449	−2.52E-04
2024–09-04 20:38:42.00	2024–09-04 20:53:42.00	15	−415.665	−0.462	−2.59E-04	2024–09-04 20:38:42.00	2024–09-04 20:53:42.00	15	−410.590	−0.456	−2.56E-04
2024–09-04 22:58:42.00	2024–09-04 23:13:42.00	15	−411.274	−0.457	−2.56E-04	2024–09-04 22:58:42.00	2024–09-04 23:13:42.00	15	−405.540	−0.451	−2.53E-04
2024–09-05 02:07:42.00	2024–09-05 02:18:42.00	11	−279.320	−0.423	−2.37E-04	2024–09-05 02:07:42.00	2024–09-05 02:18:42.00	11	−276.770	−0.419	−2.35E-04
2024–09-05 05:12:42.00	2024–09-05 05:26:42.00	14	−366.564	−0.436	−2.45E-04	2024–09-05 05:13:42.00	2024–09-05 05:26:42.00	13	−351.612	−0.451	−2.53E-04
2024–09-05 08:19:42.00	2024–09-05 08:32:42.00	13	−352.013	−0.451	−2.53E-04	2024–09-05 08:19:42.00	2024–09-05 08:32:42.00	13	−346.487	−0.444	−2.49E-04
2024–09-05 11:06:42.00	2024–09-05 11:07:42.00	1	33.820	0.564	3.16E-04	2024–09-05 11:06:42.00	2024–09-05 11:07:42.00	1	35.111	0.585	3.28E-04
2024–09-05 12:11:42.00	2024–09-05 12:26:42.00	15	−408.507	−0.454	−2.55E-04	2024–09-05 12:11:42.00	2024–09-05 12:26:42.00	15	−404.270	−0.449	−2.53E-04
2024–09-05 15:18:42.00	2024–09-05 15:33:42.00	15	−414.884	−0.461	−2.59E-04	2024–09-05 15:18:42.00	2024–09-05 15:33:42.00	15	−410.934	−0.457	−2.57E-04
2024–09-05 18:24:42.00	2024–09-05 18:39:42.00	15	−413.431	−0.459	−2.58E-04	2024–09-05 18:24:42.00	2024–09-05 18:39:42.00	15	−407.657	−0.453	−2.55E-04
2024–09-05 21:30:42.00	2024–09-05 21:46:42.00	16	−419.478	−0.437	−2.45E-04	2024–09-05 21:30:42.00	2024–09-05 21:46:42.00	16	−414.097	−0.431	−2.43E-04
2024–09-05 23:51:42.00	2024–09-06 00:06:42.00	15	−407.412	−0.453	−2.54E-04	2024–09-05 23:51:42.00	2024–09-06 00:06:42.00	15	−402.188	−0.447	−2.51E-04

References

Ahmad, N., Kilowasid, L., Fakhrurroja, H., et al., 2025. How geomagnetic storms affect the loss of Starlink satellites in February 2022? Earth, Planets Space 77, 2. https://doi.org/10.1186/s40623-024-02124-2.

Alfano, S., Oltrogge, D., Krag, H., et al., 2021. Risk assessment of recent high-interest conjunctions. Acta Astronaut. 184, 241–250. https://doi.org/10.1016/j.actaastro.2021.04.009.

Aranda, D., Gago, P., Pastor, A., et al., 2025. Drag augmentation for collision avoidance in LEO. Adv. Space Res. 75 (9), 6882–6895. https://doi.org/10.1016/j.asr.2024.07.014.

Ashurov, A.E., 2022. An effective method for detecting satellite orbital maneuvers and its application to LEO satellites. Adv. Aircr. Spacecr. Sci. 9 (4), 279–300. https://doi.org/10.12989/aas.2022.9.4.279.

Baruah, Y., Roy, S., Sinha, S., 2024. The loss of Starlink satellites in February 2022: How moderate geomagnetic storms can adversely affect assets in low-Earth orbit. Space Weather 22 (4), e2023SW003716. https://doi.org/10.1029/2023SW003716.

Burgis, S., Rohrmüller, L., Michel, M., et al., 2023. Simulation of satellites and constellations for the assessment of collision avoidance operations. CEAS Space J. 15, 655–670. https://doi.org/10.1007/s12567-022-00471-y.

Cui, Z., Xu, Y., 2022. Impact simulation of Starlink satellites on astronomical observation using worldwide telescope. Astron. Comput. 41, 100652. https://doi.org/10.1016/j.ascom.2022.100652.

Di Vruno, F., Winkel, B., Bassa, C.G., et al., 2023. Unintended electromagnetic radiation from Starlink satellites detected with LOFAR between 110 and 188 MHz. Astron. Astrophys. 676 (8), A75. https://doi.org/10.1051/0004-6361/202346374.

Dutt, P., Mutyalarao, M., Bhanumathy, P., et al., 2023. Assessment of in-house algorithms on re-entry time prediction of uncontrolled space objects. Adv. Space Res. 72 (7), 2535–2551. https://doi.org/10.1016/j.asr.2022.12.025.

Halferty, G., Reddy, V., Campbell, T., et al., 2022. Photometric characterization and trajectory accuracy of Starlink satellites: implications for ground-based astronomical surveys. Mon. Not. R. Astron. Soc. 516 (1), 1502–1508. https://doi.org/10.1093/mnras/stac2080.

Hossein, S.H., Cimino, L., Rossetti, M., et al., 2023. Photometric characterization of Starlink satellite tracklets using RGB filters. Adv. Space Res. 72 (7), 2741–2758. https://doi.org/10.1016/j.asr.2022.07.082.

Li, B., Liu, L., Sang, J.-Z., 2022. Tracklet-to-object matching for climbing Starlink satellites through recursive orbit determination and prediction. Res. Astron. Astrophys. 22 (11), 115010. https://doi.org/10.1088/1674-4527/ac91be.

Lin, C.-S., Chen, J.-Y., Zhu, H.-Z., 2025. A method for compensating prediction errors to improve the orbit prediction accuracy of Starlink satellites. GPS Solut. 29, 17. https://doi.org/10.1007/s10291-024-01771-y.

Liu, A.-R., Xiong, Y.-Q., Hui, J.-J., et al., 2024a. Research on Starlink ephemeris published by SpaceX. Acta Astron. Sin. 65 (6), 60. https://doi.org/10.15940/j.cnki.0001-5245.2024.06.005.

Liu, A.-R., Xu, X.-L., Xiong, Y.-Q., et al., 2024b. Maneuver strategies of Starlink satellite based on SpaceX-released ephemeris. Adv. Space Res. 74 (7), 3157–3169. https://doi.org/10.1016/j.asr.2024.06.038.

Liu, H., Yu, S.-X., Wang, X., 2025. Mega-constellation satellite maneuver forecast via network with attention mechanism. Adv. Space Res. 75 (6), 4942–4962. https://doi.org/10.1016/j.asr.2024.12.058.

Liu, L., 2000. Orbit Theory of Spacecraft. National Defense Industry Press, Beijing.

Liu, L., 2023. Algorithms for Satellite Orbital Dynamics. Springer, Singapore. https://doi.org/10.1007/978-981-19-4839-8.

Liu, L., Hu, S.J., Cao, J.F., et al., 2015. Theory of Orbit Determination of Spacecrafts and Application. Publishing House of Electronics Industry, Beijing.

Liu, L., Tang, J.S., 2015. Orbit Theory of Satellites and Application. Publishing House of Electronics Industry, Beijing.

Liu, L., Wang, H.H., Hu, S.J., 2005. Precise Orbit Determination of Artificial Earth Satellites. Department of Astronomy, Nanjing University.

McDowell, J., 2020. The low Earth orbit satellite population and impacts of the SpaceX Starlink constellation. Astrophys. J. Lett. 892 (2), L36. https://doi.org/10.3847/2041-8213/ab8016.

Montenbruck, O., Gill, E., 2000. Satellite Orbits: Models, Methods, and Applications. Springer, Berlin. https://doi.org/10.1007/978-3-642-58351-3.

Mróz, P., Otarola, A., Prince, T.A., et al., 2022. Impact of the SpaceX Starlink satellites on the Zwicky transient facility survey observations. Astrophys. J. Lett. 924 (2), L30. https://doi.org/10.3847/2041-8213/ac470a.

Oliveira, D.M., Zesta, E., Nandy, D., 2025. The 10 October 2024 geomagnetic storm may have caused the premature reentry of a Starlink satellite. Front. Astron. Space Sci. 11, 1522139. https://doi.org/10.3389/fspas.2024.1522139.

Pavlis, N.K., Holmes, S.A., Kenyon, S.C., et al., 2012. The development and evaluation of the Earth Gravitational Model 2008 (EGM2008). J. Geophys. Res. Solid Earth 117 (B4), B04406. https://doi.org/10.1029/2011JB008916.

Petit, A., Rossi, A., Alessi, E.M., 2021. Assessment of the close approach frequency and collision probability for satellites in different configurations of large constellations. Adv. Space Res. 67 (12), 4177–4192. https://doi.org/10.1016/j.asr.2021.02.022.

Picone, J.M., Hedin, A.E., Drob, D.P., et al., 2002. NRLMSISE-00 empirical model of the atmosphere: Statistical comparisons and scientific issues. J. Geophys. Res-Space Phys. 107 (A12), 1468. https://doi.org/10.1029/2002JA009430.

Reiland, N., Rosengren, A.J., Malhotra, R., et al., 2021. Assessing and minimizing collisions in satellite mega-constellations. Adv. Space Res. 67 (11), 3755–3774. https://doi.org/10.1016/j.asr.2021.01.010.

Shustov, B.M., 2022. Satellite mega-constellations and the dark and quiet sky problem. Astron. Rep. 66 (8), 725–735. https://doi.org/10.1134/S1063772922090104.

Tang, J.-S., Qu, Y.-Y., Wang, Q., 2023. Analysis and design of Starlink-like satellite constellation. Acta Astron. Sin. 64 (5), 52. https://doi.org/10.15940/j.cnki.0001-5245.2023.05.004.

Tyson, J., Ivezić, Z., Bradshaw, A., et al., 2020. Mitigation of LEO satellite brightness and trail effects on the Rubin observatory LSST. Astron. J. 160 (5), 226. https://doi.org/10.3847/1538-3881/abba3e.

Walker, C., Hall, J., Allen, L., et al., 2020. Impact of satellite constellations on optical astronomy and recommendations toward mitigations. Bull. Am. Astron. Soc. 52 (2). https://doi.org/10.3847/25c2cfeb.346793b8.

Wang, D.-F., Li, F., 2023. A machine learning method for the orbit state classification of large LEO constellation satellites. Adv. Space Res. 71 (3), 1644–1656. https://doi.org/10.1016/j.asr.2022.09.040.

Yuan, Y.-J., Zhang, M.-J., Wang, H.-B., et al., 2024. Retrieval and analysis of thermospheric mass densities below 200 km altitude based on precise orbit data of the re-entry objects SZ-10 MODULE and TIANGONG 1. Adv. Space Res. 73 (6), 3137–3147. https://doi.org/10.1016/j.asr.2023.12.045.

Zhang, D.-P., Shen, H.-X., 2022. Study on the orbital characteristics of Starlink satellites based on precise ephemeris. Mech. Eng. 44 (6), 1268–1278. https://doi.org/10.6052/1000-0879-22-355.

Zhang, M.-J., Wang, H.-B., Zhao, C.-Y., et al., 2018. Retrieval of atmospheric mass densities in lower thermosphere below 200 km from precise orbit of re-entry object CZ-3B R/B by analytical and numerical methods. Astrophys. Space Sci. 363 (8), 175. https://doi.org/10.1007/s10509-018-3396-9.

Zhang, W., Zhang, R.-Z., Wang, X.-H., et al., 2022. Analysis of collision threat and collision detection strategy of Starlink constellation. Manned Spaceflight 28 (5), 606–612. https://doi.org/10.16329/j.cnki.zrht.2022.05.004.